tastepocket

the pocket-conditioned finetuning set, instance by instance

343 T1 chemosensory complexes from the PDB become 269 (ligand, receptor) records, of which 243 decompose. Every record carries a 24-bit flavour vector and a 1280-d ESM-2 embedding of the 10 A pocket around its ligand.

What is in it

corpus records	243
decompositions	734 (3.02 per record)
R-groups	906 (1.23 per decomposition)
distinct ligands (CCD)	164
distinct receptors (UniProt)	98
receptor families	16
organisms	31
condvec width	1304 = 24 flavour + 1280 pocket
pocket residues (mean)	68.3
pocket sites behind them	1052

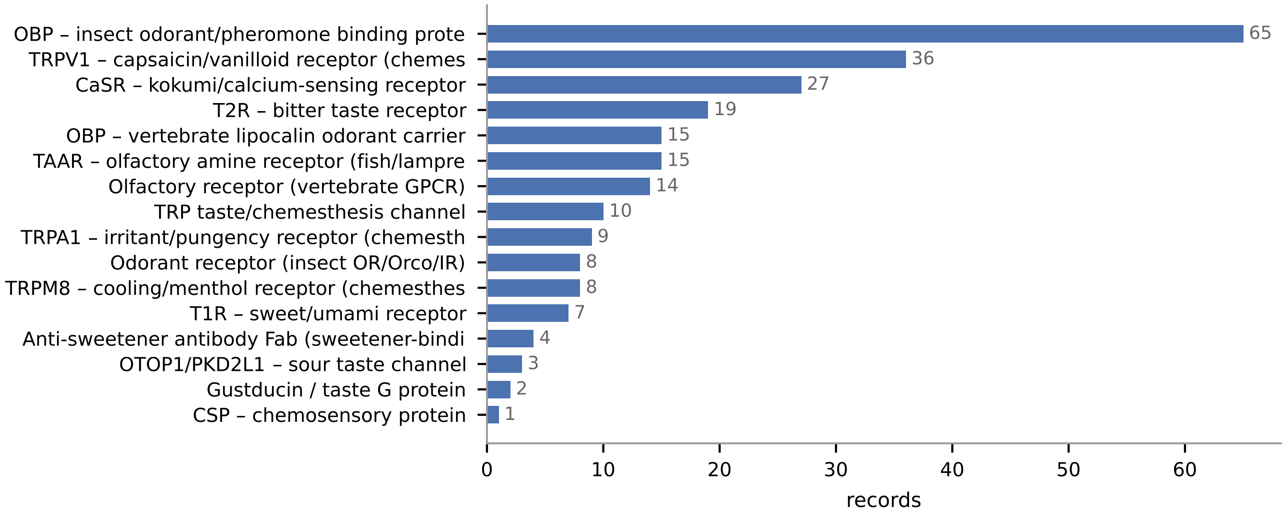
Why the counts are what they are

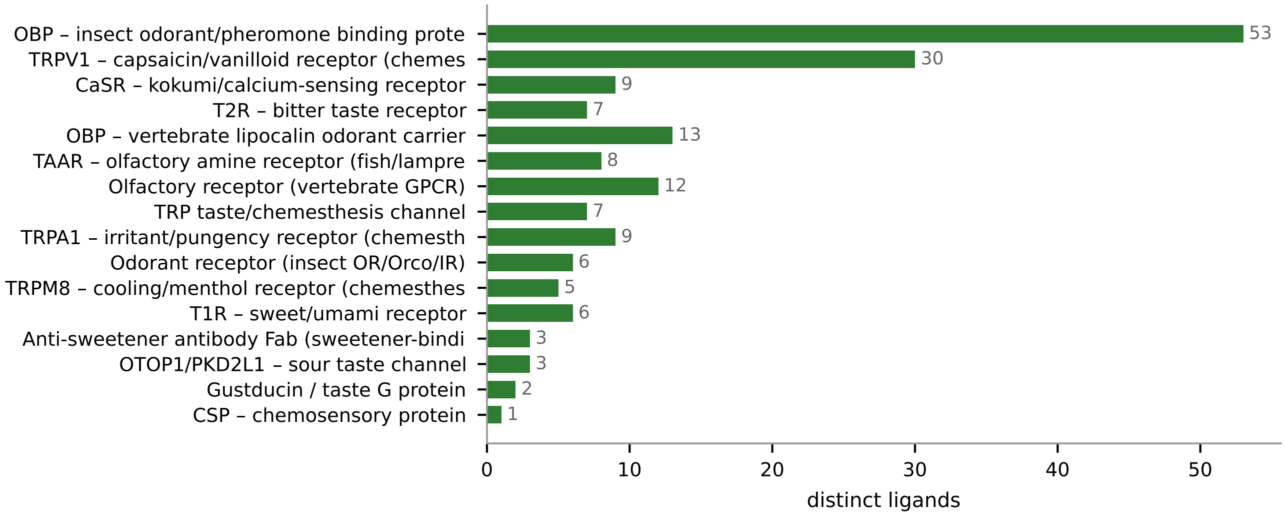
One record per (ligand, RECEPTOR), not per pocket site. 1,255 sites collapse to 269 because a homotetramer deposits the same ligand four times. Counting those separately would inflate the R-group frequency prior unevenly -- and that prior is both what Hit@K is judged against and what the logQ correction subtracts, so inflating it moves the number being optimised.

The 1,255 is itself a correction. Selecting ligand copies by residue name alone matched backbone residues too: mmCIF records free amino acids as ATOM rather than HETATM, so a `resname TRP` match swept up all 14 tryptophans in the protein. That produced 2,269 sites, 445 of them 'tryptophan'.

Receptors

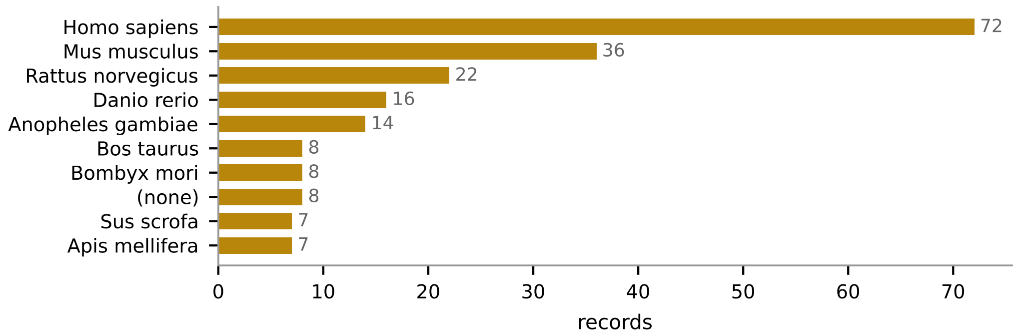
what the set actually covers





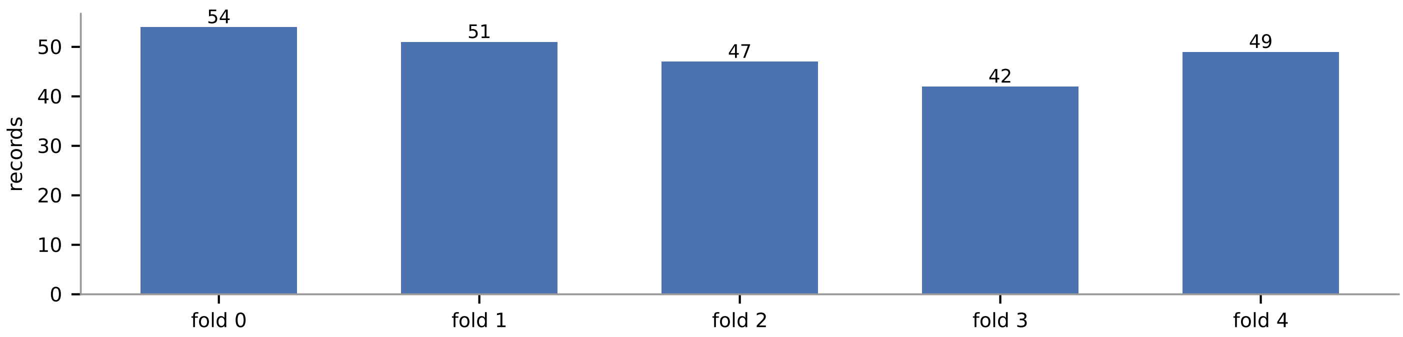
Records and ligands rank differently, and the gap is the point. Highest records-per-ligand: CaSR (3.00), T2R (2.71). Both are amino-acid or bitter sensors where the same few ligands recur across several receptors, and each (ligand, receptor) pair is a separate datapoint.

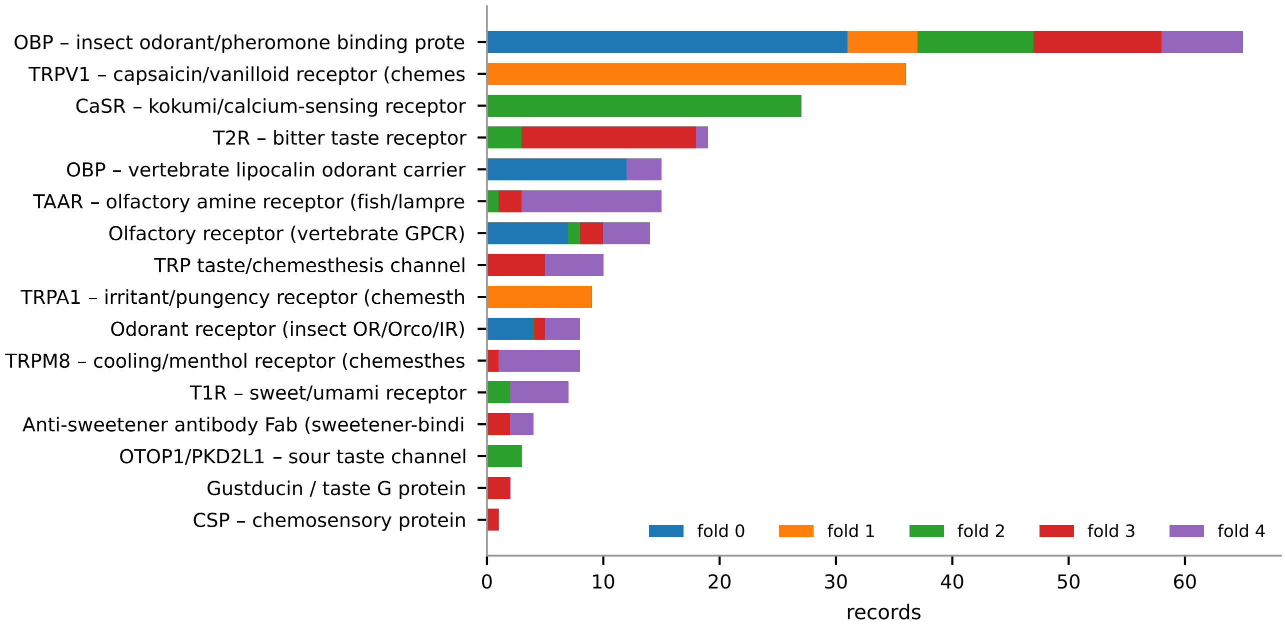
OBP is the reverse: 53 distinct ligands over 65 records, the widest chemistry in the set -- and the least labellable, since insect pheromones have no human flavour descriptor.



Cross-validation folds

cut on connected components, not at random





A fold holds out whole connected components of the (ligand, receptor) bipartite graph, so a held-out receptor's ligands are absent from training too. Splitting by receptor and patching ligand conflicts afterwards cost 64 of 269 records and still left the halves unbalanced.

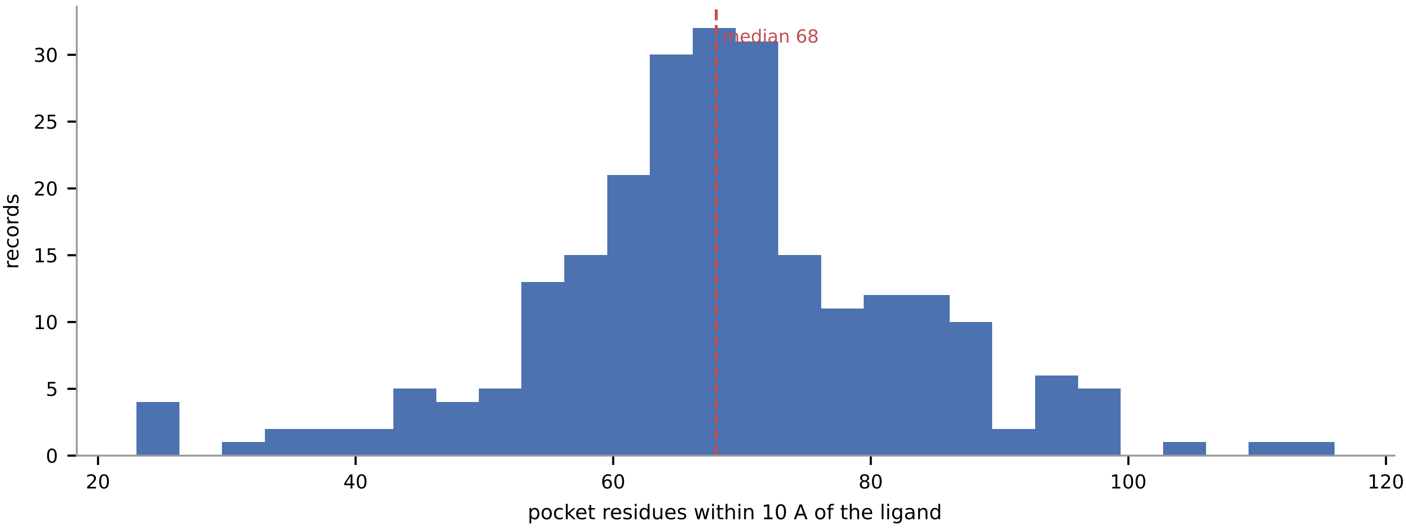
This cannot be random. Pooled ESM-2 pocket embeddings identify the receptor family with 98.5% 1-NN accuracy across different PDB entries, against a 27.4% majority baseline -- a random split puts the same protein on both sides and reports memorisation as generalisation.

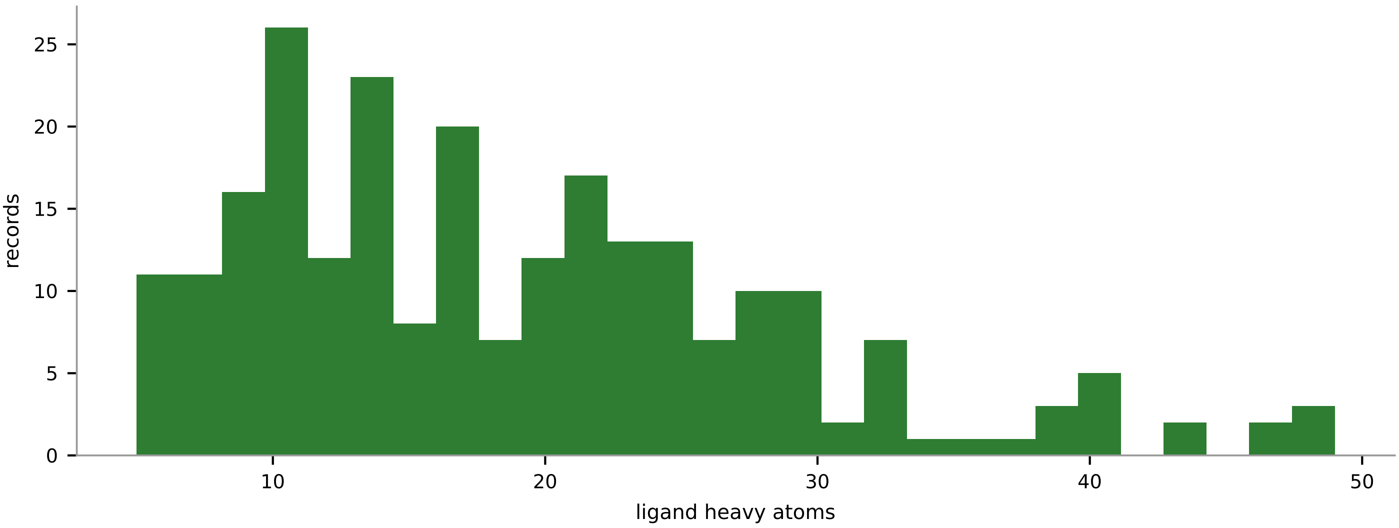
6 of 16 families sit in a single fold, because each is one component and cannot be spread at any weight. Those folds test an unseen receptor FAMILY, which is strictly harder than an unseen receptor:

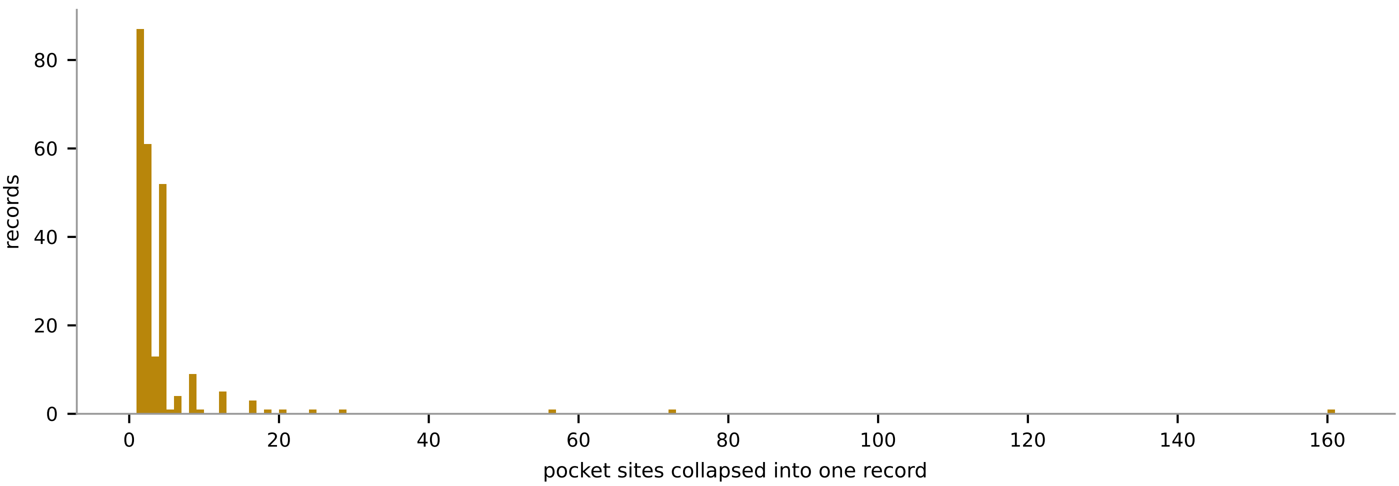
- TRPV1 – capsaicin/vanilloid receptor (chemesthesis)
- CaSR – kokumi/calcium-sensing receptor
- TRPA1 – irritant/pungency receptor (chemesthesis)
- OTOP1/PKD2L1 – sour taste channel
- Gustducin / taste G protein
- CSP – chemosensory protein

Pockets and ligands

10 A sites, and the molecules that define them



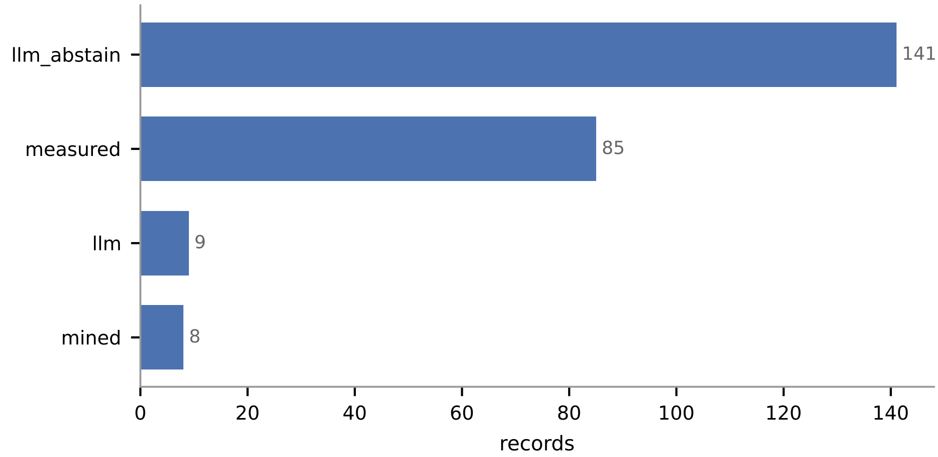


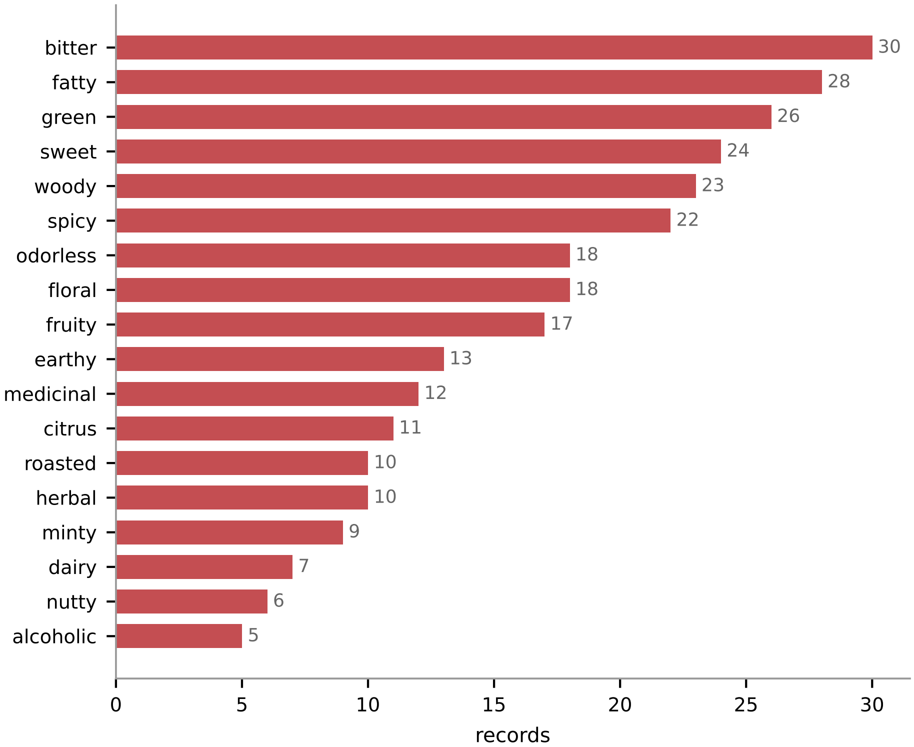


10 A matches the pocket10 convention CrossDocked's processed LMDB uses, so pockets from the two sources are directly comparable. Residue selection is 3D even though the embedding is 1D, which is what keeps it usable for GPCRs: in 8F76 the propionate site draws on residues 100-108 and 151-157.

Flavour labels

tables first, then a molecule-only LLM pass





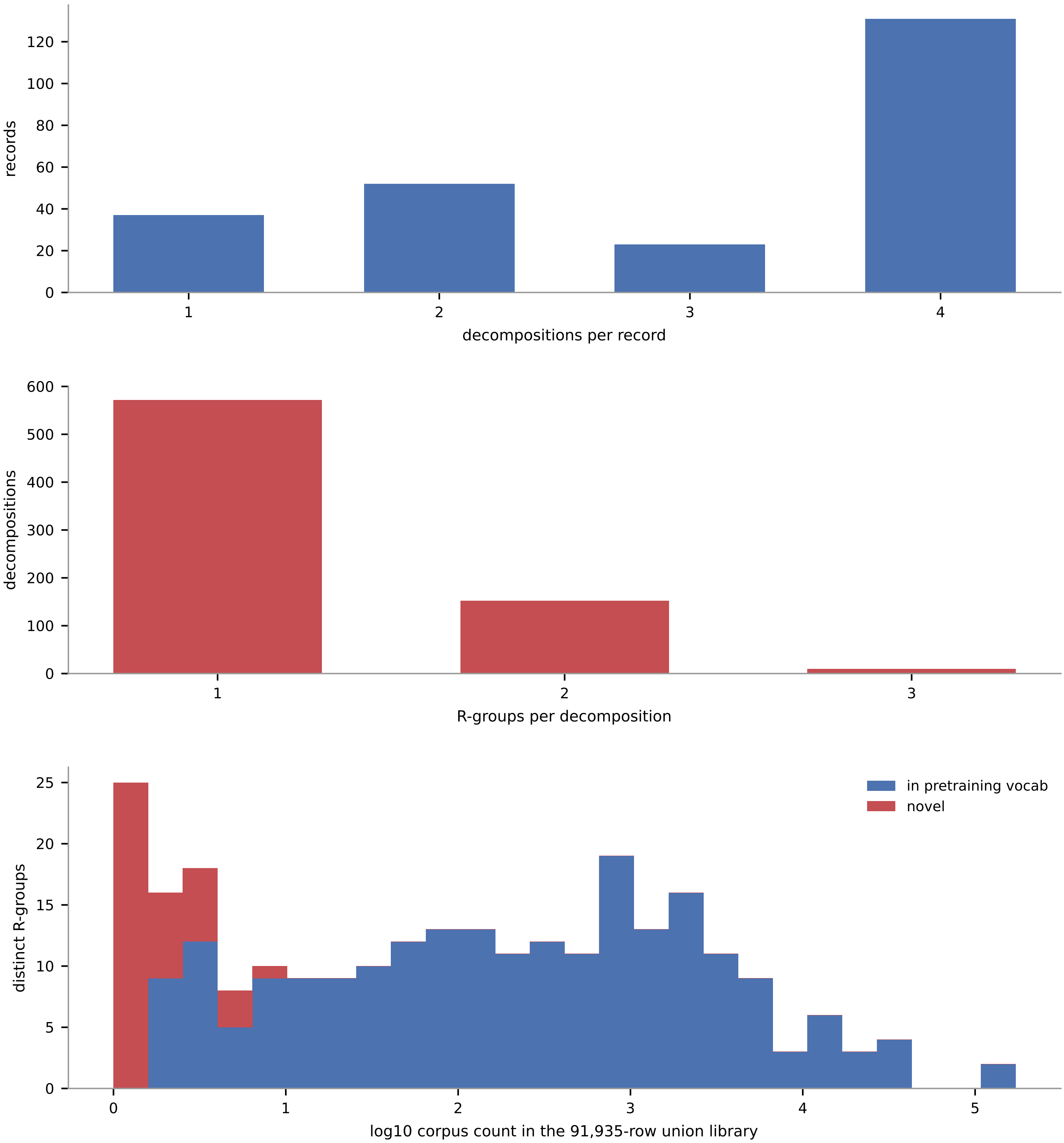
102 of 243 records (42%) carry a real label.

The rest is not a preprocessing failure. Of the molecules the tables could not resolve, most are modulators, lipids or cofactors with no percept at all, and 19 more are insect pheromones -- bombykol, bombykal, honeybee queen mandibular pheromone. Genuine stimuli with no human descriptor. Insect OBP is the largest family here, so a large part of this set is unlabellable in a 22-term human vocabulary by construction.

The LLM pass sees the MOLECULE ONLY -- name and SMILES, never the receptor. Told 'this binds TRPM8' it would answer 'cooling', and the flavour half of the condvec would become a re-encoding of the pocket half: conditioning would then show a large lift meaning nothing, because both halves would carry the same variable.

Decompositions

naveja_recap, identical settings to every other corpus



42 of 263 distinct R-groups (16%) are absent from the pretraining vocabulary. They are also RARE: median corpus count 1 against 255 for the rest.

That rarity is why hit@K has to be reported split. Novel R-groups barely move a micro average, so a comfortable headline number is consistent with novel retrieval failing outright. Whether it does is what base_hit@K and novel_hit@K are there to separate -- neither can be read off this chart.

Catalogue

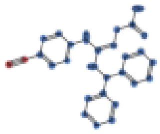
records 1–4 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

GAS · G0YP42;P01867

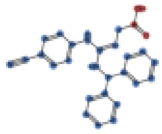
N-(P-CYANOPHENYL)-N'-DIPHENYLMETHYL-GUANIDINE-ACETIC…

```
family    Anti-sweetener antibody Fab…
organism  Mus musculus
PDB       1ETZ
fold 3   sites 2   pocket 54.0 res
flavour   (none) [llm_abstain]
SMILES    N#Cc1ccc(N/C(=N\CC(=O)O)NC(c2ccccc2)c2cc…
```

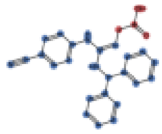
#0 C#N



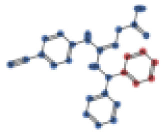
#1 O=CO



#2 CC(=O)O



#3 c1ccccc1

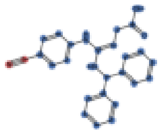


GAS · A2P1G9

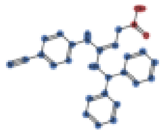
N-(P-CYANOPHENYL)-N'-DIPHENYLMETHYL-GUANIDINE-ACETIC…

```
family    Anti-sweetener antibody Fab…
organism  Mus musculus
PDB       2CGR
fold 3   sites 1   pocket 57.0 res
flavour   (none) [llm_abstain]
SMILES    N#Cc1ccc(N/C(=N\CC(=O)O)NC(c2ccccc2)c2cc…
```

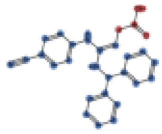
#0 C#N



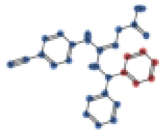
#1 O=CO



#2 CC(=O)O



#3 c1ccccc1

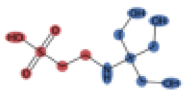


NES · A2P1G9;P01867

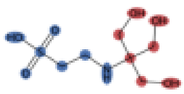
2-(2-HYDROXY-1,1-DIHYDROXYMETHYL-ETHYLAMINO)-ETHANES…

```
family    Anti-sweetener antibody Fab…
organism  Mus musculus
PDB       1YNL
fold 4   sites 1   pocket 55.0 res
flavour   (none) [llm_abstain]
SMILES    O=S(=O)(O)CCNC(CO)(CO)CO
```

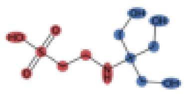
#0 CCS(=O)(=O)O



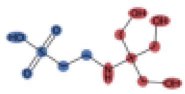
#1 OCC(CO)CO



#2 NCCS(=O)(=O)O



#3 NC(CO)(CO)CO

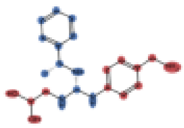


SC5 · A2P1G9;P01867

2-[((R)-{[4-(AMINOMETHYL)PHENYL]AMINO}{[(1R)-1-PHENY…

```
family    Anti-sweetener antibody Fab…
organism  Mus musculus
PDB       1YNK
fold 4   sites 1   pocket 58.0 res
flavour   sweet [mined]
SMILES    C[C@@H](N[C@@H](NCC(O)O)Nc1ccc(CN)cc1)c1…
```

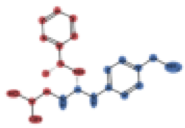
#0 CC(O)O…



#1 NCc1ccc(N)cc1…

#2 NCC(O)O…

#3 C[C@@H](N)c1cccc…



Catalogue

records 5–8 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

BDD · Q9NG96

BROMO-DODECANOL

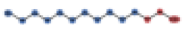
```
family    CSP — chemosensory protein
organism  Mamestra brassicae
PDB       1N8U 1N8V
fold 3   sites 9   pocket 67.2 res
flavour   (none) [llm_abstain]
SMILES    OCCCCCCCCCCCCBr
```

#0 CBr

#1 CO

#2 CCBr

#3 CCO

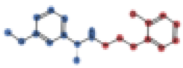


9IG · P41180

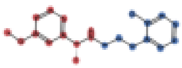
3-(2-chlorophenyl)-N-[(1R)-1-(3-methoxyphenyl)ethyl]...

```
family    CaSR — kokumi/calcium-sensing receptor
organism  Homo sapiens
PDB       7SIL
fold 2   sites 2   pocket 68.0 res
flavour   (none) [llm_abstain]
SMILES    COc1cccc([C@@H](C)NCCCc2ccccc2Cl)c1
```

#0 CCCc1ccccc1Cl ●



#1 COc1cccc([C@@H](... ●



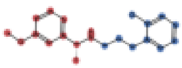
9IG · A0A590UJY2;P41180;P59768;P6287…

3-(2-chlorophenyl)-N-[(1R)-1-(3-methoxyphenyl)ethyl]...

```
family    CaSR — kokumi/calcium-sensing receptor
organism  Homo sapiens
PDB       9ASB
fold 2   sites 2   pocket 71.0 res
flavour   (none) [llm_abstain]
SMILES    COc1cccc([C@@H](C)NCCCc2ccccc2Cl)c1
```

#0 CCCc1ccccc1Cl ●

#1 COc1cccc([C@@H](... ●



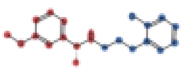
9IG · P41180;P59768;P62873;P63092

3-(2-chlorophenyl)-N-[(1R)-1-(3-methoxyphenyl)ethyl]...

```
family    CaSR — kokumi/calcium-sensing receptor
organism  Homo sapiens
PDB       9AXF
fold 2   sites 2   pocket 67.5 res
flavour   (none) [llm_abstain]
SMILES    COc1cccc([C@@H](C)NCCCc2ccccc2Cl)c1
```

#0 CCCc1ccccc1Cl ●

#1 COc1cccc([C@@H](... ●



Catalogue

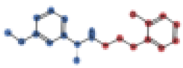
records 9–12 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

9IG · P41180;P59768;P62873;P63092;P6…

3-(2-chlorophenyl)-N-[(1R)-1-(3-methoxyphenyl)ethyl]…

family CaSR — kokumi/calcium-sensing receptor
organism Homo sapiens
PDB 9AVG
fold 2 sites 2 pocket 72.0 res
flavour (none) [llm_abstain]
SMILES COc1cccc([C@@H](C)NCCCc2ccccc2Cl)c1

#0 CCCc1ccccc1Cl ● #1 COc1cccc([C@@H](… ●

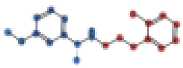



9IG · P08754;P41180;P59768;P62879

3-(2-chlorophenyl)-N-[(1R)-1-(3-methoxyphenyl)ethyl]…

family CaSR — kokumi/calcium-sensing receptor
organism Homo sapiens
PDB 9AVL
fold 2 sites 2 pocket 67.0 res
flavour (none) [llm_abstain]
SMILES COc1cccc([C@@H](C)NCCCc2ccccc2Cl)c1

#0 CCCc1ccccc1Cl ● #1 COc1cccc([C@@H](… ●

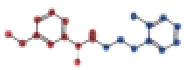



9IG · P41180;P59768;P62873;P63096

3-(2-chlorophenyl)-N-[(1R)-1-(3-methoxyphenyl)ethyl]…

family CaSR — kokumi/calcium-sensing receptor
organism Homo sapiens
PDB 9AYF
fold 2 sites 2 pocket 64.5 res
flavour (none) [llm_abstain]
SMILES COc1cccc([C@@H](C)NCCCc2ccccc2Cl)c1

#0 CCCc1ccccc1Cl ● #1 COc1cccc([C@@H](… ●

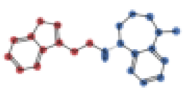
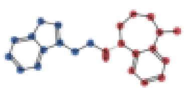



A1ATP · P41180

(5R)-N-[2-(1,2-benzothiazol-3-yl)ethyl]-1-methyl-2,3…

family CaSR — kokumi/calcium-sensing receptor
organism Homo sapiens
PDB 9C1P
fold 2 sites 2 pocket 62.5 res
flavour (none) [llm_abstain]
SMILES CN1CCC[C@@H](NCCc2nsc3ccccc23)c2ccccc21

#0 CCc1nsc2ccccc12 ● #1 CN1CCC[C@@H](N)c… ●

Catalogue

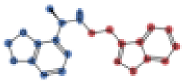
records 13–16 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

A1ATX · P41180

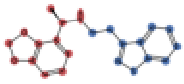
(1R)-1-(2H-1,3-benzodioxol-4-yl)-N-[2-(1,2-benzothia…

```
family    CaSR — kokumi/calcium-sensing receptor
organism  Homo sapiens
PDB       9C2F
fold 2   sites 2   pocket 65.5 res
flavour   (none) [llm_abstain]
SMILES    C[C@@H](NCCc1nsc2ccccc12)c1cccc2c1OCO2
```

#0 CCc1nsc2ccccc12 ●



#1 C[C@@H](N)c1cccc… ●

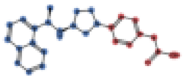


H43 · P41180

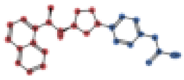
2-[4-[(3S)-3-[[(1R)-1-naphthalen-1-ylethyl]amino]pyr…

```
family    CaSR — kokumi/calcium-sensing receptor
organism  Homo sapiens
PDB       7M3G
fold 2   sites 2   pocket 67.0 res
flavour   (none) [llm_abstain]
SMILES    C[C@@H](N[C@H]1CCN(c2ccc(CC(=O)O)cc2)C1)…
```

#0 O=C(O)Cc1ccccc1



#1 C[C@@H](N[C@H]1C…



SPM · P41180

SPERMINE

```
family    CaSR — kokumi/calcium-sensing receptor
organism  Homo sapiens
PDB       8SZF
fold 2   sites 1   pocket 47.0 res
flavour   (none) [llm_abstain]
SMILES    NCCCNCCCCNCCCN
```

#0 CCCN CCCN

#1 CCCN NCCCN

#2 NCCCN CCCN

#3 CCCN

SPM · P08754;P41180;P59768;P62873

SPERMINE

```
family    CaSR — kokumi/calcium-sensing receptor
organism  Homo sapiens
PDB       8SZH
fold 2   sites 4   pocket 37.5 res
flavour   (none) [llm_abstain]
SMILES    NCCCNCCCCNCCCN
```

#0 CCCN CCCN

#1 CCCN NCCCN

#2 NCCCN CCCN

#3 CCCN

Catalogue

records 17–20 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

SPM · P41180;P50148;P59768;P62873;P6…

SPERMINE

```
family    CaSR — kokumi/calcium-sensing receptor
organism  Homo sapiens
PDB       8SZG
fold 2   sites 3   pocket 35.0 res
flavour   (none) [llm_abstain]
SMILES    NCCCNCCCCNCCCN
```

#0 CCCN CCCN

#1 CCCN NCCCN

#2 NCCCN CCCN

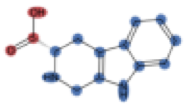
#3 CCCN

TCR · P41180;P59768;P62873;P63096

CYCLOMETHYLTRYPTOPHAN

```
family    CaSR — kokumi/calcium-sensing receptor
organism  Homo sapiens
PDB       9AYF
fold 2   sites 2   pocket 63.5 res
flavour   (none) [llm_abstain]
SMILES    O=C(O)[C@@H]1Cc2c([nH]c3ccccc23)CN1
```

#0 O=CO

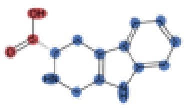


TCR · A0A590UJY2;P41180;P59768;P6287…

CYCLOMETHYLTRYPTOPHAN

```
family    CaSR — kokumi/calcium-sensing receptor
organism  Homo sapiens
PDB       9ASB
fold 2   sites 2   pocket 70.0 res
flavour   (none) [llm_abstain]
SMILES    O=C(O)[C@@H]1Cc2c([nH]c3ccccc23)CN1
```

#0 O=CO

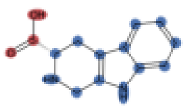


TCR · P41180;P59768;P62873;P63092;P6…

CYCLOMETHYLTRYPTOPHAN

```
family    CaSR — kokumi/calcium-sensing receptor
organism  Homo sapiens
PDB       9AVG
fold 2   sites 2   pocket 71.0 res
flavour   (none) [llm_abstain]
SMILES    O=C(O)[C@@H]1Cc2c([nH]c3ccccc23)CN1
```

#0 O=CO



Catalogue

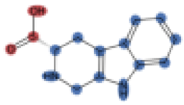
records 21–24 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

TCR · P41180

CYCLOMETHYLTRYPTOPHAN

```
family    CaSR — kokumi/calcium-sensing receptor
organism  Homo sapiens
PDB       7E6T 7SIL 7SIM
fold 2   sites 6   pocket 67.3 res
flavour   (none) [llm_abstain]
SMILES    O=C(O)[C@@H]1Cc2c([nH]c3ccccc23)CN1
```

#0 O=CO

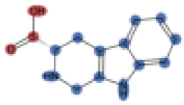


TCR · P08754;P41180;P59768;P62879

CYCLOMETHYLTRYPTOPHAN

```
family    CaSR — kokumi/calcium-sensing receptor
organism  Homo sapiens
PDB       9AVL
fold 2   sites 2   pocket 66.5 res
flavour   (none) [llm_abstain]
SMILES    O=C(O)[C@@H]1Cc2c([nH]c3ccccc23)CN1
```

#0 O=CO

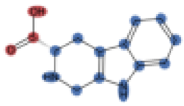


TCR · P41180;P59768;P62873;P63092

CYCLOMETHYLTRYPTOPHAN

```
family    CaSR — kokumi/calcium-sensing receptor
organism  Homo sapiens
PDB       9AXF
fold 2   sites 2   pocket 65.0 res
flavour   (none) [llm_abstain]
SMILES    O=C(O)[C@@H]1Cc2c([nH]c3ccccc23)CN1
```

#0 O=CO

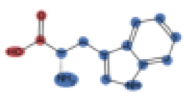


TRP · P41180;P50148;P59768;P62873;P6…

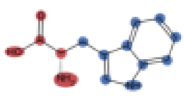
TRYPTOPHAN

```
family    CaSR — kokumi/calcium-sensing receptor
organism  Homo sapiens
PDB       8SZG
fold 2   sites 2   pocket 67.5 res
flavour   bitter [measured]
SMILES    N[C@@H](Cc1c[nH]c2ccccc12)C(=O)O
```

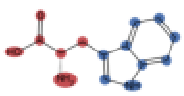
#0 O=CO



#1 NCC(=O)O



#2 C[C@H](N)C(=O)O



Catalogue

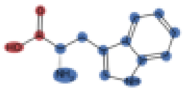
records 25–28 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

TRP · P08754;P41180;P59768;P62873

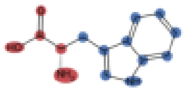
TRYPTOPHAN

```
family    CaSR — kokumi/calcium-sensing receptor
organism  Homo sapiens
PDB       8SZH 8SZI
fold 2   sites 5   pocket 64.4 res
flavour   bitter [measured]
SMILES    N[C@@H](Cc1c[nH]c2ccccc12)C(=O)O
```

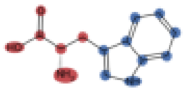
#0 O=CO



#1 NCC(=O)O



#2 C[C@H](N)C(=O)O

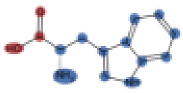


TRP · P41180

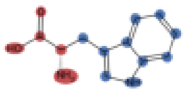
TRYPTOPHAN

```
family    CaSR — kokumi/calcium-sensing receptor
organism  Homo sapiens
PDB       7DTU 7DTV 7M3E 7M3F 7M3G 8SZF 8WPG 9C1P…
fold 2   sites 18   pocket 67.2 res
flavour   bitter [measured]
SMILES    N[C@@H](Cc1c[nH]c2ccccc12)C(=O)O
```

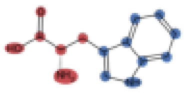
#0 O=CO



#1 NCC(=O)O



#2 C[C@H](N)C(=O)O

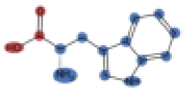


TRP · P04899;P41180;P59768;P62873;P6…

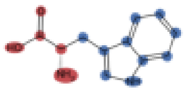
TRYPTOPHAN

```
family    CaSR — kokumi/calcium-sensing receptor
organism  Homo sapiens
PDB       8WPU
fold 2   sites 2   pocket 71.5 res
flavour   bitter [measured]
SMILES    N[C@@H](Cc1c[nH]c2ccccc12)C(=O)O
```

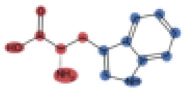
#0 O=CO



#1 NCC(=O)O



#2 C[C@H](N)C(=O)O

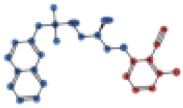


YP1 · P41180

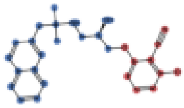
2-chloro-6-[(2R)-2-hydroxy-3-{[2-methyl-1-(naphthale…

```
family    CaSR — kokumi/calcium-sensing receptor
organism  Homo sapiens
PDB       7M3E 7M3J 7SIN
fold 2   sites 6   pocket 68.5 res
flavour   (none) [llm_abstain]
SMILES    CC(C)(Cc1ccc2ccccc2c1)NC[C@@H](O)COc1ccc…
```

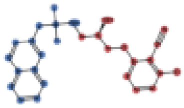
#0 N#Cc1ccccc1Cl



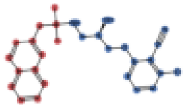
#1 N#Cc1c(O)cccc1Cl



#2 C[C@@H](O)COc1cc…



#3 CC(C)Cc1ccc2cccc…



Catalogue

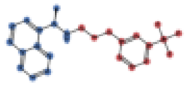
records 29–32 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

YP4 · P08754;P41180;P59768;P62873

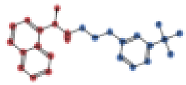
N-[(1R)-1-(naphthalen-1-yl)ethyl]-3-[3-(trifluoromet…

```
family    CaSR — kokumi/calcium-sensing receptor
organism  Homo sapiens
PDB       8SZH
fold 2   sites 2   pocket 61.5 res
flavour   (none) [llm_abstain]
SMILES    C[C@@H](NCCCc1cccc(C(F)(F)F)c1)c1cccc2cc…
```

#0 CCCc1cccc(C(F)(F…



● #1 C[C@@H](N)c1cccc…

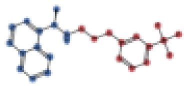


YP4 · P04899;P41180;P59768;P62873;P6…

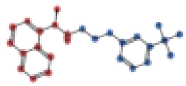
N-[(1R)-1-(naphthalen-1-yl)ethyl]-3-[3-(trifluoromet…

```
family    CaSR — kokumi/calcium-sensing receptor
organism  Homo sapiens
PDB       8WPU
fold 2   sites 2   pocket 69.0 res
flavour   (none) [llm_abstain]
SMILES    C[C@@H](NCCCc1cccc(C(F)(F)F)c1)c1cccc2cc…
```

#0 CCCc1cccc(C(F)(F…



● #1 C[C@@H](N)c1cccc…

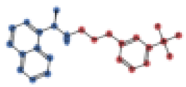


YP4 · P41180;P50148;P59768;P62873;P6…

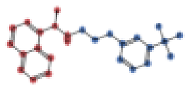
N-[(1R)-1-(naphthalen-1-yl)ethyl]-3-[3-(trifluoromet…

```
family    CaSR — kokumi/calcium-sensing receptor
organism  Homo sapiens
PDB       8SZG
fold 2   sites 2   pocket 64.0 res
flavour   (none) [llm_abstain]
SMILES    C[C@@H](NCCCc1cccc(C(F)(F)F)c1)c1cccc2cc…
```

#0 CCCc1cccc(C(F)(F…



● #1 C[C@@H](N)c1cccc…

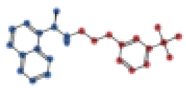


YP4 · P41180

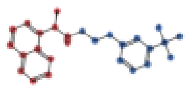
N-[(1R)-1-(naphthalen-1-yl)ethyl]-3-[3-(trifluoromet…

```
family    CaSR — kokumi/calcium-sensing receptor
organism  Homo sapiens
PDB       7M3F 8SZF 8WPG
fold 2   sites 6   pocket 64.5 res
flavour   (none) [llm_abstain]
SMILES    C[C@@H](NCCCc1cccc(C(F)(F)F)c1)c1cccc2cc…
```

#0 CCCc1cccc(C(F)(F…



● #1 C[C@@H](N)c1cccc…



Catalogue

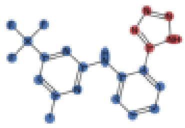
records 33–36 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

A1AEI · A8MTJ3;P59768;P62873;Q59310;Q9…

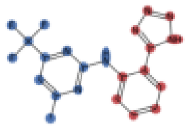
4-methyl-N-[(2M)-2-(1H-tetrazol-5-yl)phenyl]-6-(trif…

```
family    Gustducin / taste G protein
organism  Homo sapiens
PDB       8YKY
fold 3   sites 1   pocket 72.0 res
flavour   (none) [llm_abstain]
SMILES    Cc1cc(C(F)(F)F)nc(Nc2ccccc2-c2nnn[nH]2)n…
```

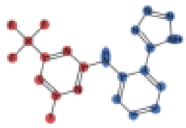
#0 c1nnn[nH]1



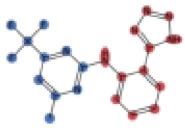
#1 c1ccc(-c2nnn[nH]…



#2 Cc1cc(C(F)(F)F)n… ●



#3 Nc1ccccc1-c1nnn[…

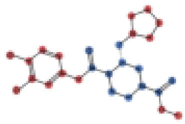


U9I · P41145;P59768;P62873

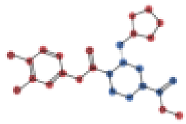
methyl…

```
family    Gustducin / taste G protein
organism  Homo sapiens
PDB       8DZR
fold 3   sites 1   pocket 88.0 res
flavour   (none) [llm_abstain]
SMILES    COC(=O)N1CCN(C(=O)Cc2ccc(Cl)c(Cl)c2)[C@H…
```

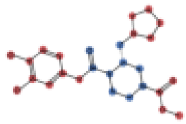
#0 CO C1CCNC1…



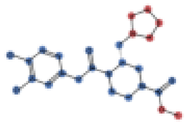
#1 CO…



#2 COC=O C1CCNC1…



#3 CO C1CCNC1

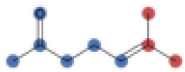


0VT · Q8I8T0

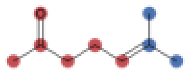
6-methylhept-5-en-2-one

```
family    OBP — insect odorant/pheromone binding…
organism  Anopheles gambiae
PDB       4FQT
fold 1   sites 2   pocket 60.0 res
flavour   bitter;citrus;earthy;fruity;gr… [measured]
SMILES    CC(=O)CCC=C(C)C
```

#0 CCC



#1 CCCC(C)=O

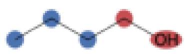


1BO · O02372

1-BUTANOL

```
family    OBP — insect odorant/pheromone binding…
organism  Drosophila melanogaster
PDB       1OOH 3B6X
fold 0   sites 4   pocket 56.5 res
flavour   alcoholic;fatty;fruity;medicin… [measured]
SMILES    CCCCO
```

#0 CO



#1 CC

Catalogue

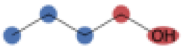
records 37–40 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

1BO · Q8T6R6

1-BUTANOL

```
family    OBP — insect odorant/pheromone binding…
organism  Anopheles gambiae
PDB       8BXW
fold 0   sites 1   pocket 23.0 res
flavour   alcoholic;fatty;fruity;medicin… [measured]
SMILES    CCCCO
```

#0 CO



#1 CC

1EX · D0E9M1

(11Z,13Z)-hexadeca-11,13-dien-1-ol

```
family    OBP — insect odorant/pheromone binding…
organism  Amyelois transitella
PDB       4INX
fold 3   sites 1   pocket 87.0 res
flavour   (none) [llm_abstain]
SMILES    CC/C=C\C=C/CCCCCCCCCCO
```

#0 CCC

#1 ?

1EY · D0E9M1

(11Z,13Z)-hexadeca-11,13-dienal

```
family    OBP — insect odorant/pheromone binding…
organism  Amyelois transitella
PDB       4INW
fold 3   sites 1   pocket 88.0 res
flavour   (none) [llm_abstain]
SMILES    CC/C=C\C=C/CCCCCCCCCC=O
```

#0 CCC

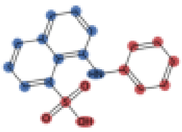
#1 ?

2AN · Q8MTC1

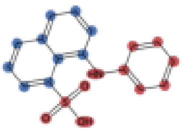
8-ANILINO-1-NAPHTHALENE SULFONATE

```
family    OBP — insect odorant/pheromone binding…
organism  Rhyparobia maderae
PDB       1OW4
fold 1   sites 2   pocket 78.5 res
flavour   (none) [llm_abstain]
SMILES    O=S(=O)(O)c1cccc2cccc(Nc3ccccc3)c12
```

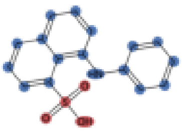
#0 O=S(=O)O…



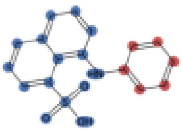
#1 O=S(=O)O…



#2 O=S(=O)O



#3 c1ccccc1



Catalogue

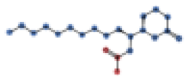
records 41–44 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

3OG · Q8T6I2

(1S)-1-[(2R)-6-oxotetrahydro-2H-pyran-2-yl]undecyl…

```
family    OBP — insect odorant/pheromone binding…
organism  Culex quinquefasciatus
PDB       3OGN
fold 4   sites 2   pocket 82.5 res
flavour   (none) [llm_abstain]
SMILES    CCCCCCCCCC[C@H](OC(C)=O)[C@H]1CCCC(=O)O1
```

#0 CC=O



3OL · P07435

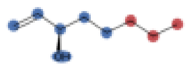
(3R)-oct-1-en-3-ol

```
family    OBP — insect odorant/pheromone binding…
organism  Bos taurus
PDB       1G85
fold 0   sites 2   pocket 65.5 res
flavour   earthy;fatty;green;meaty;odorl… [measured]
SMILES    C=C[C@H](O)CCCCC
```

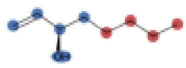
#0 CC

#1 C=C

#2 CCC



#3 CCCC

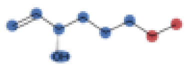


3OM · P07435

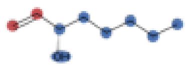
(3S)-1-octen-3-ol

```
family    OBP — insect odorant/pheromone binding…
organism  Bos taurus
PDB       1GT1 1GT3
fold 0   sites 3   pocket 65.0 res
flavour   earthy;fatty;green;meaty;odorl… [measured]
SMILES    C=C[C@@H](O)CCCCC
```

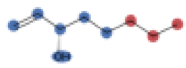
#0 CC



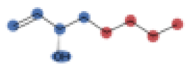
#1 C=C



#2 CCC



#3 CCCC



81K · F5ANH9

(Z)-hexadec-9-enal

```
family    OBP — insect odorant/pheromone binding…
organism  Helicoverpa armigera
PDB       7VWA
fold 2   sites 1   pocket 76.0 res
flavour   (none) [llm_abstain]
SMILES    CCCCCC/C=C\CCCCCCCC=O
```

#0 CCCCCCC

#1 CCCCCCC=O

#2 C=C\CCCCCC

#3 CCCCCCCC=O

Catalogue

records 45–48 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

9OD · Q9U9J6

(2Z)-9-oxodec-2-enoic acid

```
family    OBP — insect odorant/pheromone binding…
organism  Apis mellifera
PDB       3BFA 3BFB 3CYZ 3CZ0
fold 2   sites 8   pocket 67.6 res
flavour   (none) [llm_abstain]
SMILES    CC(=O)CCCCC/C=C\C(=O)O
```

#0 CC=O CC(=O)O

#1 CC(C)=O CC(=O)O

#2 CC=O

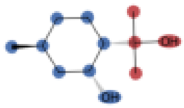
#3 CC(=O)O

A1J26 · Q8I8R2

P-Menthane-3,8-diol

```
family    OBP — insect odorant/pheromone binding…
organism  Anopheles gambiae
PDB       29MD
fold 3   sites 1   pocket 61.0 res
flavour   herbal;minty [measured]
SMILES    C[C@H]1CC[C@H](C(C)(C)O)[C@H](O)C1
```

#0 CC(C)O

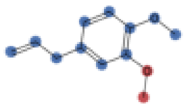


A1J27 · Q8I8R2

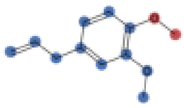
Methyleugenol

```
family    OBP — insect odorant/pheromone binding…
organism  Anopheles gambiae
PDB       29MC
fold 3   sites 1   pocket 66.0 res
flavour   bitter;dairy;earthy;fatty;frui… [measured]
SMILES    C=CCc1ccc(OC)c(OC)c1
```

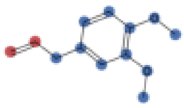
#0 CO



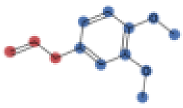
#1 CO



#2 C=C



#3 C=CC

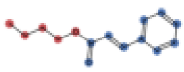


A1J28 · Q8I8R2

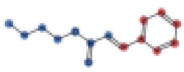
Butyl Cinnamate

```
family    OBP — insect odorant/pheromone binding…
organism  Anopheles gambiae
PDB       29LY
fold 3   sites 1   pocket 75.0 res
flavour   fruity;roasted;spicy;sweet;woo… [measured]
SMILES    CCCCOC(=O)/C=C/c1ccccc1
```

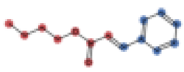
#0 CCCCO



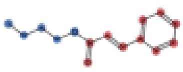
#1 Cc1ccccc1



#2 CC(=O)OCCCC



#3 O=C/C=C/c1ccccc1



Catalogue

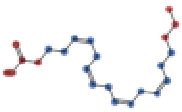
records 49–52 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

ACD · Q1HRL7

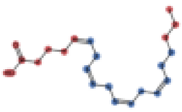
ARACHIDONIC ACID

```
family    OBP — insect odorant/pheromone binding…
organism  Aedes aegypti
PDB       6NBN 6OII
fold 3   sites 3   pocket 98.7 res
flavour   (none) [llm_abstain]
SMILES    CCCCC/C=C\C/C=C\C/C=C\C/C=C\CCCC(=O)O
```

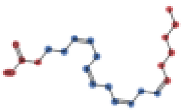
#0 CCC CC(=O)O



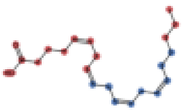
#1 CCC CCCCC(=O)O



#2 CCCCCC CC(=O)O



#3 CCC...



AZB · Q7PXT9

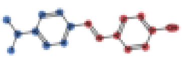
4-{(E)-[4-(propan-2-yl)phenyl]diazenyl}phenol

```
family    OBP — insect odorant/pheromone binding…
organism  Anopheles gambiae
PDB       3R1V
fold 2   sites 2   pocket 76.5 res
flavour   (none) [llm_abstain]
SMILES    CC(C)c1ccc(/N=N/c2ccc(O)cc2)cc1
```

#0 CCC

#1 Oc1ccccc1

#2 N=N/c1ccc(O)cc1



B7M · P34170

(10E,12Z)-hexadeca-10,12-dienal

```
family    OBP — insect odorant/pheromone binding…
organism  Bombyx mori
PDB       2WCH
fold 0   sites 1   pocket 81.0 res
flavour   (none) [llm_abstain]
SMILES    CCC/C=C\C=C\CCCCCCCCC=O
```

#0 CCCC CCCC=O

#1 CCCC CCCCC=O

#2 CCCC CCCCCC=O

#3 ? CCCC=O

B8M · P34170

(8E,10Z)-HEXADECA-8,10-DIEN-1-OL

```
family    OBP — insect odorant/pheromone binding…
organism  Bombyx mori
PDB       2WCL
fold 0   sites 1   pocket 85.0 res
flavour   (none) [llm_abstain]
SMILES    CCCCC/C=C\C=C\CCCCCCCO
```

#0 CCCCCC

#1 CCCCCCO

#2 CCCCCCCO

#3 ?

Catalogue

records 53–56 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

B9M · P34170

(10E)-hexadec-10-en-12-yn-1-ol

family   OBP — insect odorant/pheromone binding…
organism Bombyx mori
PDB   2WCM
fold 0   sites 1   pocket 82.0 res
flavour   (none) [llm_abstain]
SMILES   CCCC#C/C=C/CCCCCCCCCO

#0 CC#CCCC

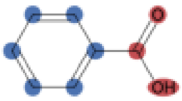
#1 CCCCCCCCCCO

BEZ · Q1HRL7

BENZOIC ACID

family   OBP — insect odorant/pheromone binding…
organism Aedes aegypti
PDB   6OTL 6P2E
fold 3   sites 2   pocket 48.0 res
flavour   odorless;woody [measured]
SMILES   O=C(O)c1ccccc1

#0 O=CO



BOM · P34174

HEXADECA-10,12-DIEN-1-OL

family   OBP — insect odorant/pheromone binding…
organism Bombyx mori
PDB   1DQE
fold 0   sites 2   pocket 84.5 res
flavour   (none) [llm_abstain]
SMILES   CCC/C=C\C=C\CCCCCCCCCO

#0 CCCC CCCCO

#1 CCCC CCCCCO

#2 CCCC CCCCCCO

#3 ? CCCCO

BOM · P34170

HEXADECA-10,12-DIEN-1-OL

family   OBP — insect odorant/pheromone binding…
organism Bombyx mori
PDB   2WC5 2WC6
fold 0   sites 2   pocket 85.0 res
flavour   (none) [llm_abstain]
SMILES   CCC/C=C\C=C\CCCCCCCCCO

#0 CCCC CCCCO

#1 CCCC CCCCCO

#2 CCCC CCCCCCO

#3 ? CCCCO

Catalogue

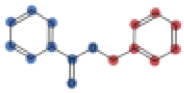
records 57–60 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

BZM · P81245

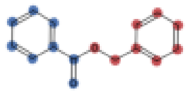
BENZOIC ACID PHENYLMETHYLESTER

```
family    OBP — insect odorant/pheromone binding…
organism  Sus scrofa
PDB       1DZM
fold 0   sites 2   pocket 73.0 res
flavour   dairy;fatty;fruity;herbal;nutt… [measured]
SMILES    O=C(OCc1ccccc1)c1ccccc1
```

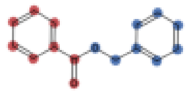
#0 Cc1ccccc1



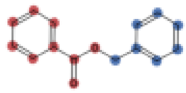
#1 OCc1ccccc1



#2 O=Cc1ccccc1



#3 O=C(O)c1ccccc1

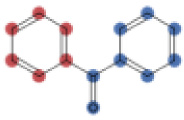


BZQ · P81245

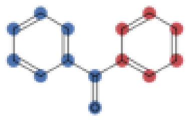
DIPHENYLMETHANONE

```
family    OBP — insect odorant/pheromone binding…
organism  Sus scrofa
PDB       1DZP
fold 0   sites 2   pocket 68.5 res
flavour   floral;fruity;medicinal;woody [measured]
SMILES    O=C(c1ccccc1)c1ccccc1
```

#0 c1ccccc1



#1 c1ccccc1

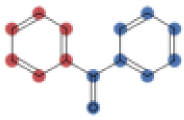


BZQ · P07435

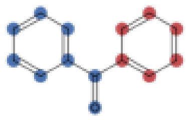
DIPHENYLMETHANONE

```
family    OBP — insect odorant/pheromone binding…
organism  Bos taurus
PDB       1GT5
fold 0   sites 2   pocket 70.0 res
flavour   floral;fruity;medicinal;woody [measured]
SMILES    O=C(c1ccccc1)c1ccccc1
```

#0 c1ccccc1



#1 c1ccccc1

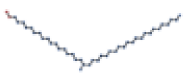


CMJ · Q9U9J6

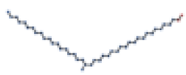
(20S)-20-methyldotetracontane

```
family    OBP — insect odorant/pheromone binding…
organism  Apis mellifera
PDB       3FE6 3FE8 3FE9
fold 2   sites 3   pocket 85.7 res
flavour   (none) [llm_abstain]
SMILES    CCCCCCCCCCCCCCCCCCCCCC[C@@H](C)CCCCCCCCC…
```

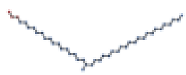
#0 CC



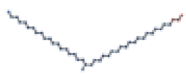
#1 CC



#2 CCC



#3 CCC



Catalogue

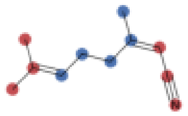
records 61–64 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

CTV · Q1W640

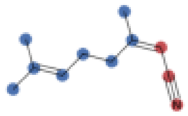
(2Z)-3,7-dimethylocta-2,6-dienenitrile

```
family    OBP — insect odorant/pheromone binding…
organism  Apis mellifera
PDB       3S0D
fold 4   sites 1   pocket 66.0 res
flavour   citrus;green;fruity [llm]
SMILES    CC(C)=CCC/C(C)=C\C#N
```

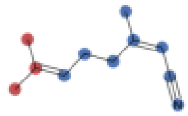
#0 CCC CC#N



#1 CC#N



#2 CCC



DCR · Q1HRL7

icosanoic acid

```
family    OBP — insect odorant/pheromone binding…
organism  Aedes aegypti
PDB       6OPB
fold 3   sites 3   pocket 97.0 res
flavour   odorless [mined]
SMILES    CCCCCCCCCCCCCCCCCCCC(=O)O
```

#0 CC

#1 O=CO

#2 CCC

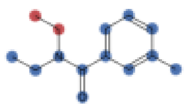
#3 CC(=O)O

DE3 · Q8T6S0

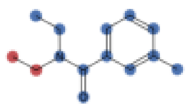
N,N-diethyl-3-methylbenzamide

```
family    OBP — insect odorant/pheromone binding…
organism  Anopheles gambiae
PDB       3N7H
fold 4   sites 2   pocket 57.0 res
flavour   (none) [llm_abstain]
SMILES    CCN(CC)C(=O)c1cccc(C)c1
```

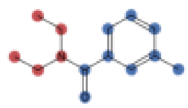
#0 CC



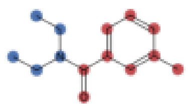
#1 CC



#2 CCNCC



#3 Cc1cccc(C=O)c1

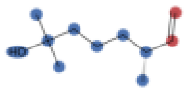


DHM · P07435

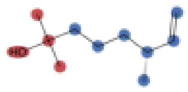
2,6-DIMETHYL-7-OCTEN-2-OL

```
family    OBP — insect odorant/pheromone binding…
organism  Bos taurus
PDB       1GT3
fold 0   sites 2   pocket 66.0 res
flavour   citrus;floral;green;odorless [measured]
SMILES    C=C[C@H](C)CCCC(C)(C)O
```

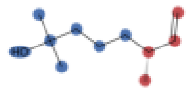
#0 C=C



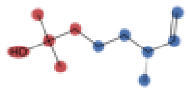
#1 CC(C)O



#2 C=CCC



#3 CC(C)(C)O



Catalogue

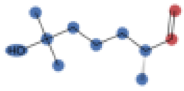
records 65–68 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

DHM · P81245

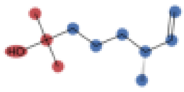
2,6-DIMETHYL-7-OCTEN-2-OL

```
family    OBP — insect odorant/pheromone binding…
organism  Sus scrofa
PDB       1E00
fold 0   sites 2   pocket 72.5 res
flavour   citrus;floral;green;odorless [measured]
SMILES    C=C[C@H](C)CCCC(C)(C)O
```

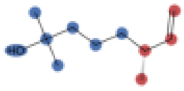
#0 C=C



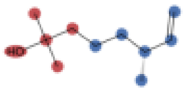
#1 CC(C)O



#2 C=CCC



#3 CC(C)(C)O

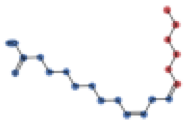


EIC · Q1HRL7

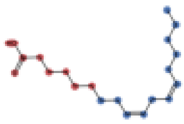
LINOLEIC ACID

```
family    OBP — insect odorant/pheromone binding…
organism  Aedes aegypti
PDB       6OGH
fold 3   sites 1   pocket 95.0 res
flavour   fatty;odorless [measured]
SMILES    CCCCC/C=C\C/C=C\CCCCCCCC(=O)O
```

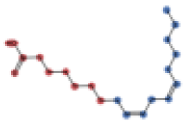
#0 CCCCCC



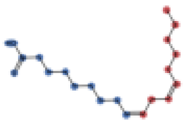
#1 CCCCCC(=O)O



#2 CCCCCCC(=O)O



#3 CC/C=C\CCCCC

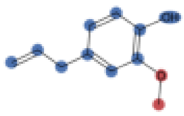


EOL · Q1W640

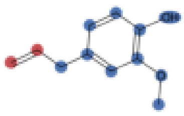
2-methoxy-4-(prop-2-en-1-yl)phenol

```
family    OBP — insect odorant/pheromone binding…
organism  Apis mellifera
PDB       3S0E
fold 4   sites 1   pocket 65.0 res
flavour   spicy;sweet;woody [measured]
SMILES    C=CCc1ccc(O)c(OC)c1
```

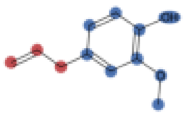
#0 CO



#1 C=C



#2 C=CC



ETE · O02372

2-{2-[2-2-(METHOXY-ETHOXY)-ETHOXY]-ETHOXY}-ETHANOL

```
family    OBP — insect odorant/pheromone binding…
organism  Drosophila melanogaster
PDB       3B88
fold 0   sites 2   pocket 62.0 res
flavour   (none) [llm_abstain]
SMILES    COCCOCCOCCOCCO
```

#0 CO CCO

#1 CO OCCO

#2 CCOC CCO

#3 CO CCOCCO

Catalogue

records 69–72 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

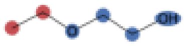
ETX · Q8T6R6

2-ETHOXYETHANOL

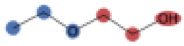
```
family    OBP — insect odorant/pheromone binding…
organism  Anopheles gambiae
PDB       8BXU
fold 0   sites 1   pocket 26.0 res
flavour   (none) [llm_abstain]
SMILES    CCOCCO
```

#0 CO

#1 CC



#2 CCO

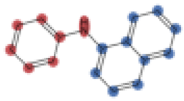


FNA · Q1W640

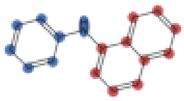
N-phenylnaphthalen-1-amine

```
family    OBP — insect odorant/pheromone binding…
organism  Apis mellifera
PDB       3S0B
fold 4   sites 1   pocket 72.0 res
flavour   (none) [llm_abstain]
SMILES    c1ccc(Nc2cccc3ccccc23)cc1
```

#0 Nc1ccccc1



#1 c1ccc2ccccc2c1

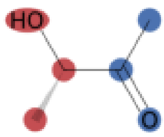


HBS · Q8MTC1

S,3-HYDROXYBUTAN-2-ONE

```
family    OBP — insect odorant/pheromone binding…
organism  Rhyparobia maderae
PDB       1P28
fold 1   sites 2   pocket 59.0 res
flavour   dairy;fatty;sweet [measured]
SMILES    CC(=O)[C@H](C)O
```

#0 CCO

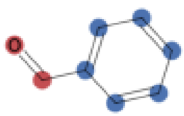


HBX · Q7PGA3

benzaldehyde

```
family    OBP — insect odorant/pheromone binding…
organism  Anopheles gambiae
PDB       3L4L
fold 4   sites 1   pocket 54.0 res
flavour   bitter;fruity;nutty;spicy;swee… [measured]
SMILES    O=Cc1ccccc1
```

#0 C=O



Catalogue

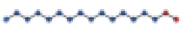
records 73–76 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

IHD · P34174

1-IODOHEXADECANE

```
family    OBP — insect odorant/pheromone binding…
organism  Bombyx mori
PDB       2P71
fold 0    sites 1   pocket 83.0 res
flavour   (none) [llm_abstain]
SMILES    CCCCCCCCCCCCCCCCI
```

#0 CI



#1 CC

#2 CCI

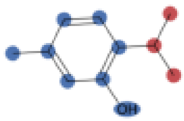
#3 CCC

IPB · P81245

5-METHYL-2-(1-METHYLETHYL)PHENOL

```
family    OBP — insect odorant/pheromone binding…
organism  Sus scrofa
PDB       1E06
fold 0    sites 2   pocket 66.5 res
flavour   herbal;medicinal;woody [measured]
SMILES    Cc1ccc(C(C)C)c(O)c1
```

#0 CCC

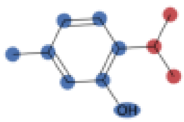


IPB · Q8T6R6

5-METHYL-2-(1-METHYLETHYL)PHENOL

```
family    OBP — insect odorant/pheromone binding…
organism  Anopheles gambiae
PDB       8BXV
fold 0    sites 1   pocket 65.0 res
flavour   herbal;medicinal;woody [measured]
SMILES    Cc1ccc(C(C)C)c(O)c1
```

#0 CCC



JH3 · Q16Y94

methyl…

```
family    OBP — insect odorant/pheromone binding…
organism  Aedes aegypti
PDB       5V13
fold 1    sites 3   pocket 84.3 res
flavour   (none) [llm_abstain]
SMILES    COC(=O)/C=C(\C)CC/C=C(\C)CC[C@H]1OC1(C)C
```

#0 CO…

#1 CO…

#2 CC(=O)OC…

#3 CO

Catalogue

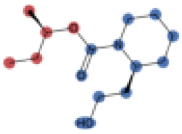
records 77–80 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

KBR · Q8I8T0

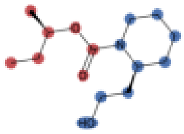
Icaridin

```
family    OBP — insect odorant/pheromone binding…
organism  Anopheles gambiae
PDB       5EL2
fold 1   sites 4   pocket 60.5 res
flavour   (none) [llm_abstain]
SMILES    CC[C@@H](C)OC(=O)N1CCCC[C@H]1CCO
```

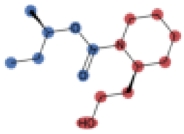
#0 CC[C@@H](C)O



#1 CC[C@@H](C)OC=O



#2 OCC[C@@H]1CCCCN1

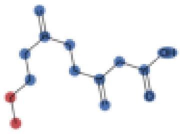


LIK · P07435

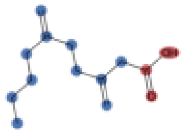
3,6-BIS(METHYLENE)DECANOIC ACID

```
family    OBP — insect odorant/pheromone binding…
organism  Bos taurus
PDB       2HLV
fold 0   sites 1   pocket 72.0 res
flavour   (none) [llm_abstain]
SMILES    C=C(CCCC)CCC(=C)CC(=O)O
```

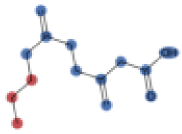
#0 CC



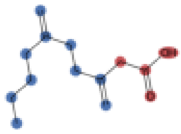
#1 O=CO



#2 CCC



#3 CC(=O)O

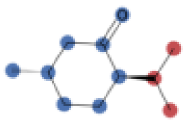


LWE · H2B3G5

(+)-menthone

```
family    OBP — insect odorant/pheromone binding…
organism  Canis lupus familiaris
PDB       8AEJ
fold 3   sites 1   pocket 65.0 res
flavour   minty [measured]
SMILES    CC(C)[C@H]1CC[C@H](C)CC1=O
```

#0 CCC



M21 · P34170

(10E,12Z)-tetradeca-10,12-dien-1-ol

```
family    OBP — insect odorant/pheromone binding…
organism  Bombyx mori
PDB       2WCJ
fold 0   sites 1   pocket 81.0 res
flavour   (none) [llm_abstain]
SMILES    C/C=C\C=C\CCCCCCCCCO
```

#0 CC

#1 ?

Catalogue

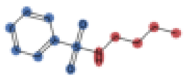
records 81–84 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

NBB · Q8WRW2

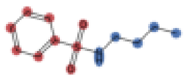
N-BUTYL-BENZENESULFONAMIDE

```
family    OBP — insect odorant/pheromone binding…
organism  Apis mellifera
PDB       3R72
fold 2   sites 2   pocket 63.5 res
flavour   (none) [llm_abstain]
SMILES    CCCCNS(=O)(=O)c1ccccc1
```

#0 CCCCN



#1 O=S(=O)c1ccccc1

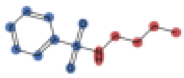


NBB · Q9U9J6

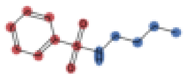
N-BUTYL-BENZENESULFONAMIDE

```
family    OBP — insect odorant/pheromone binding…
organism  Apis mellifera
PDB       3BJH 3CZ1 3D73 3D74 3D75 3D76 3D77 3D78
fold 2   sites 16   pocket 62.2 res
flavour   (none) [llm_abstain]
SMILES    CCCCNS(=O)(=O)c1ccccc1
```

#0 CCCCN



#1 O=S(=O)c1ccccc1

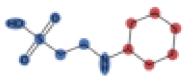


NHE · A0A7M7KVA6

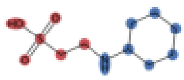
2-[N-CYCLOHEXYLAMINO]ETHANE SULFONIC ACID

```
family    OBP — insect odorant/pheromone binding…
organism  Varroa destructor
PDB       7NZA
fold 2   sites 2   pocket 56.0 res
flavour   (none) [llm_abstain]
SMILES    O=S(=O)(O)CCNC1CCCCC1
```

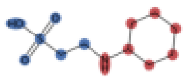
#0 C1CCCCC1



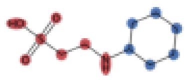
#1 CCS(=O)(=O)O



#2 NC1CCCCC1



#3 NCCS(=O)(=O)O

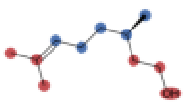


ODM · O18874

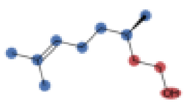
(3R)-3,7-dimethyloct-6-en-1-ol

```
family    OBP — insect odorant/pheromone binding…
organism  Canis lupus familiaris
PDB       8AEI
fold 4   sites 2   pocket 76.5 res
flavour   bitter;fatty;floral;green [measured]
SMILES    CC(C)=CCC[C@@H](C)CCO
```

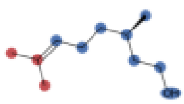
#0 CCC CCO



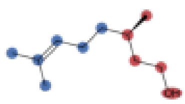
#1 CCO



#2 CCC



#3 CCCCO



Catalogue

records 85–88 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

PAM · Q1HRL7

PALMITOLEIC ACID

```
family    OBP — insect odorant/pheromone binding…
organism  Aedes aegypti
PDB       6OMW
fold 3   sites 3   pocket 94.0 res
flavour   (none) [llm_abstain]
SMILES    CCCCCC/C=C\CCCCCCCC(=O)O
```

#0 CCCCCCC

#1 C=C\CCCCCC

#2 CCCCCCC(=O)O

#3 C/C=C\CCCCCC

PG0 · W8W3V2

2-(2-METHOXYETHOXY)ETHANOL

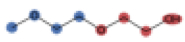
```
family    OBP — insect odorant/pheromone binding…
organism  Ceratitis capitata
PDB       6HHE
fold 2   sites 1   pocket 23.0 res
flavour   (none) [llm_abstain]
SMILES    COCCOCCO
```

#0 CO CCO

#1 CO

#2 CCO

#3 OCCO



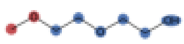
PG0 · A0A0K8TU48

2-(2-METHOXYETHOXY)ETHANOL

```
family    OBP — insect odorant/pheromone binding…
organism  Epiphyas postvittana
PDB       6VQ5
fold 2   sites 2   pocket 30.5 res
flavour   (none) [llm_abstain]
SMILES    COCCOCCO
```

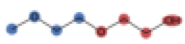
#0 CO CCO

#1 CO



#2 CCO

#3 OCCO



PG0 · Q3HM32

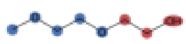
2-(2-METHOXYETHOXY)ETHANOL

```
family    OBP — insect odorant/pheromone binding…
organism  Locusta migratoria
PDB       4PT1
fold 2   sites 4   pocket 57.0 res
flavour   (none) [llm_abstain]
SMILES    COCCOCCO
```

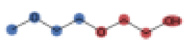
#0 CO CCO

#1 CO

#2 CCO



#3 OCCO



Catalogue

records 89–92 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

PG6 · Q7Q9J3

1-(2-METHOXY-ETHOXY)-2-{2-[2-(2-METHOXY-ETHOXY]-ETHO…

```
family    OBP — insect odorant/pheromone binding…
organism  Anopheles gambiae
PDB       4F7F
fold 1   sites 3   pocket 84.3 res
flavour   (none) [llm_abstain]
SMILES    COCCOCCOCCOCCOCCOC
```

#0 CO CO

#1 CO CCOC

#2 CCOC CO

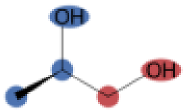
#3 CO COCCO

PGR · Q8T6R6

R-1,2-PROPANEDIOL

```
family    OBP — insect odorant/pheromone binding…
organism  Anopheles gambiae
PDB       8BXV 8BXW
fold 0   sites 2   pocket 26.0 res
flavour   alcoholic;odorless [measured]
SMILES    C[C@@H](O)CO
```

#0 CO

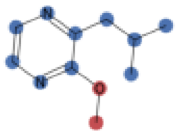


PRZ · P34174

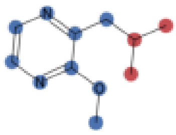
2-ISOBUTYL-3-METHOXYPYRAZINE

```
family    OBP — insect odorant/pheromone binding…
organism  Bombyx mori
PDB       2P70
fold 0   sites 2   pocket 66.0 res
flavour   earthy;green;spicy [measured]
SMILES    COc1nccnc1CC(C)C
```

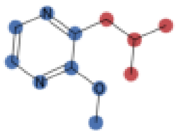
#0 CO



#1 CCC



#2 CC(C)C

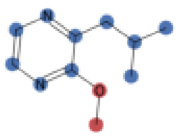


PRZ · P07435

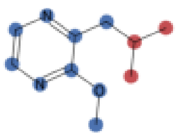
2-ISOBUTYL-3-METHOXYPYRAZINE

```
family    OBP — insect odorant/pheromone binding…
organism  Bos taurus
PDB       1GT1
fold 0   sites 1   pocket 71.0 res
flavour   earthy;green;spicy [measured]
SMILES    COc1nccnc1CC(C)C
```

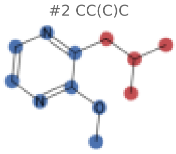
#0 CO



#1 CCC



#2 CC(C)C



Catalogue

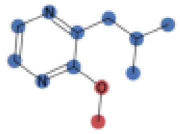
records 93–96 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

PRZ · P81245

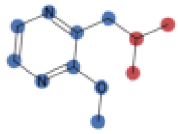
2-ISOBUTYL-3-METHOXYPYRAZINE

```
family    OBP — insect odorant/pheromone binding…
organism  Sus scrofa
PDB       1DZK 1HQP
fold 0   sites 3   pocket 68.0 res
flavour   earthy;green;spicy [measured]
SMILES    COc1nccnc1CC(C)C
```

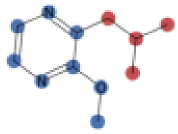
#0 CO



#1 CCC



#2 CC(C)C

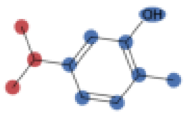


S5V · Q8T6R6

2-methyl-5-propan-2-yl-phenol

```
family    OBP — insect odorant/pheromone binding…
organism  Anopheles gambiae
PDB       8BXW
fold 0   sites 2   pocket 60.0 res
flavour   spicy;woody [measured]
SMILES    Cc1ccc(C(C)C)cc1O
```

#0 CCC

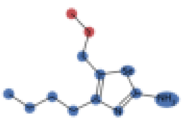


SES · P07435

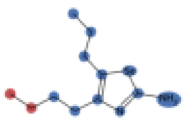
4-butyl-5-propyl-1,3-selenazol-2-amine

```
family    OBP — insect odorant/pheromone binding…
organism  Bos taurus
PDB       1PBO
fold 0   sites 2   pocket 69.5 res
flavour   (none) [llm_abstain]
SMILES    CCCCc1nc(N)[se]c1CCC
```

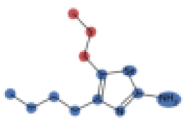
#0 CC



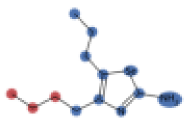
#1 CC



#2 CCC



#3 CCC

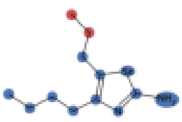


SES · P81245

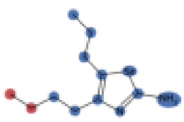
4-butyl-5-propyl-1,3-selenazol-2-amine

```
family    OBP — insect odorant/pheromone binding…
organism  Sus scrofa
PDB       1DZJ
fold 0   sites 2   pocket 71.5 res
flavour   (none) [llm_abstain]
SMILES    CCCCc1nc(N)[se]c1CCC
```

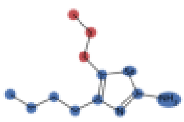
#0 CC



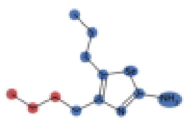
#1 CC



#2 CCC



#3 CCC



Catalogue

records 97–100 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

UNA · P07435

UNDECANAL

family OBP — insect odorant/pheromone binding…
organism Bos taurus
PDB 1GT4
fold 0 sites 2 pocket 65.0 res
flavour citrus;fatty;floral;green;spic… [measured]
SMILES CCCCCCCCCCC=O

#0 C=O

#1 CC

#2 CC=O

#3 CCC

UNA · P81245

UNDECANAL

family OBP — insect odorant/pheromone binding…
organism Sus scrofa
PDB 1E02
fold 0 sites 2 pocket 69.0 res
flavour citrus;fatty;floral;green;spic… [measured]
SMILES CCCCCCCCCCC=O

#0 C=O

#1 CC

#2 CC=O

#3 CCC

VA · O02372

(Z)-OCTADEC-11-ENYL ACETATE

family OBP — insect odorant/pheromone binding…
organism Drosophila melanogaster
PDB 2GTE
fold 0 sites 2 pocket 88.0 res
flavour (none) [llm_abstain]
SMILES CCCCCC/C=C\CCCCCCCCCCOC(C)=O

#0 CCCCCCC CC=O

#1 CCCCCCC CC(=O)O

#2 C=C\CCCCCC CC=O

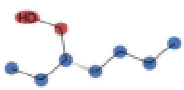
#3 C=C\CCCCCC...

2EH · P11590

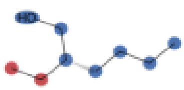
(2S)-2-ethylhexan-1-ol

family OBP — vertebrate lipocalin odorant…
organism Mus musculus
PDB 3KFH
fold 4 sites 1 pocket 61.0 res
flavour citrus;fatty;floral;green;swee… [measured]
SMILES CCCC[C@H](CC)CO

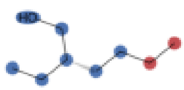
#0 CO



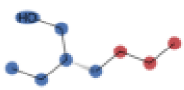
#1 CC



#2 CC



#3 CCC



Catalogue

records 101–104 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

DE1 · P11589

DECAN-1-OL

```
family    OBP — vertebrate lipocalin odorant…
organism  Mus musculus
PDB       1ZNL
fold 0   sites 1   pocket 77.0 res
flavour   citrus;fatty;floral;odorless;s… [measured]
SMILES    CCCCCCCCCCO
```

#0 CO

#1 CC

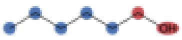
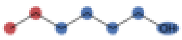
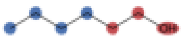
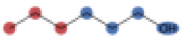
#2 CCO

#3 CCC


HE2 · P11589

HEXAN-1-OL

```
family    OBP — vertebrate lipocalin odorant…
organism  Mus musculus
PDB       1ZNE
fold 0   sites 1   pocket 60.0 res
flavour   alcoholic;fatty;floral;fruity;… [measured]
SMILES    CCCCCCO
```

#0 CO

#1 CC

#2 CCO

#3 CCC


HE4 · P11589

HEPTAN-1-OL

```
family    OBP — vertebrate lipocalin odorant…
organism  Mus musculus
PDB       1ZNG
fold 0   sites 1   pocket 72.0 res
flavour   earthy;floral;fruity;green;her… [measured]
SMILES    CCCCCCCO
```

#0 CO

#1 CC

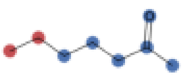
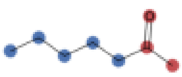
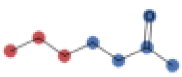
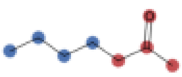
#2 CCO

#3 CCC


HTX · P11590

heptan-2-one

```
family    OBP — vertebrate lipocalin odorant…
organism  Mus musculus
PDB       3KFG
fold 4   sites 1   pocket 59.0 res
flavour   fatty;fruity;herbal;nutty;spic… [measured]
SMILES    CCCCCC(C)=O
```

#0 CC

#1 CC=O

#2 CCC

#3 CC(C)=O


Catalogue

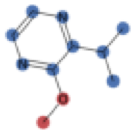
records 105–108 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

IPZ · P11589

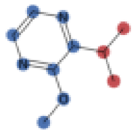
2-ISOPROPYL-3-METHOXYPYRAZINE

```
family    OBP — vertebrate lipocalin odorant…
organism  Mus musculus
PDB       1QY2
fold 0   sites 1   pocket 63.0 res
flavour   earthy [measured]
SMILES    COc1nccnc1C(C)C
```

#0 CO



#1 CCC

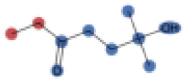


LTL · P02762

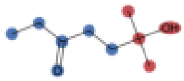
6-HYDROXY-6-METHYL-HEPTAN-3-ONE

```
family    OBP — vertebrate lipocalin odorant…
organism  Mus musculus
PDB       1I05
fold 0   sites 1   pocket 67.0 res
flavour   (none) [llm_abstain]
SMILES    CCC(=O)CCC(C)(C)O
```

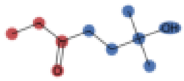
#0 CC



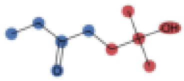
#1 CC(C)O



#2 CCC=O



#3 CC(C)(C)O



OC9 · P11589

OCTAN-1-OL

```
family    OBP — vertebrate lipocalin odorant…
organism  Mus musculus
PDB       1ZNH
fold 0   sites 1   pocket 75.0 res
flavour   citrus;earthy;fatty;floral;gre… [measured]
SMILES    CCCCCCCCO
```

#0 CO

#1 CC

#2 CCO

#3 CCC

ODI · P11588

OCTANE-1,8-DIOL

```
family    OBP — vertebrate lipocalin odorant…
organism  Mus musculus
PDB       2DM5
fold 0   sites 1   pocket 77.0 res
flavour   (none) [llm_abstain]
SMILES    OCCCCCCCCO
```

#0 CO

#1 CO

#2 CCO

#3 CCO

Catalogue

records 109–112 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

PE9 · P11589

PENTAN-1-OL

```
family    OBP — vertebrate lipocalin odorant…
organism  Mus musculus
PDB       1ZND
fold 0   sites 2   pocket 59.5 res
flavour   alcoholic;fatty;sweet;woody [measured]
SMILES    CCCCCO
```

#0 CO

#1 CC

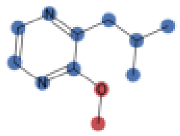
#2 CCO

PRZ · P11588

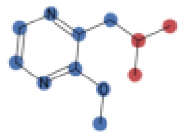
2-ISOBUTYL-3-METHOXYPYRAZINE

```
family    OBP — vertebrate lipocalin odorant…
organism  Mus musculus
PDB       1QY1 1YP6
fold 0   sites 2   pocket 70.0 res
flavour   earthy;green;spicy [measured]
SMILES    COc1nccnc1CC(C)C
```

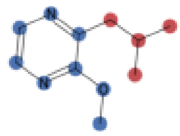
#0 CO



#1 CCC



#2 CC(C)C

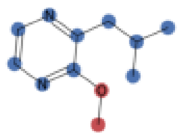


PRZ · P11589

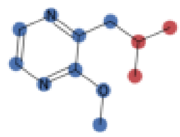
2-ISOBUTYL-3-METHOXYPYRAZINE

```
family    OBP — vertebrate lipocalin odorant…
organism  Mus musculus
PDB       2NND
fold 0   sites 1   pocket 69.0 res
flavour   earthy;green;spicy [measured]
SMILES    COc1nccnc1CC(C)C
```

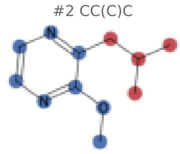
#0 CO



#1 CCC



#2 CC(C)C

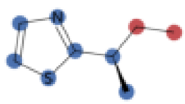


TZL · P11589

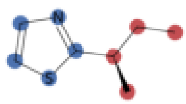
2-(SEC-BUTYL)THIAZOLE

```
family    OBP — vertebrate lipocalin odorant…
organism  Mus musculus
PDB       1JV4
fold 0   sites 1   pocket 67.0 res
flavour   green;herbal;odorless;spicy [measured]
SMILES    CC[C@@H](C)c1nccs1
```

#0 CC



#1 CCCC



Catalogue

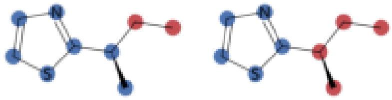
records 113–116 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

TZL · P02762

2-(SEC-BUTYL)THIAZOLE

```
family    OBP — vertebrate lipocalin odorant…
organism  Mus musculus
PDB       1I06 1MUP
fold 0   sites 2   pocket 64.5 res
flavour   green;herbal;odorless;spicy [measured]
SMILES    CC[C@@H](C)c1nccs1
```

#0 CC   #1 CCCC

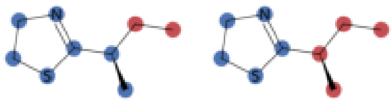


XBT · P11590

2-[(1R)-1-methylpropyl]-4,5-dihydro-1,3-thiazole

```
family    OBP — vertebrate lipocalin odorant…
organism  Mus musculus
PDB       3KFF
fold 4   sites 1   pocket 63.0 res
flavour   sulfurous [llm]
SMILES    CC[C@@H](C)C1=NCCS1
```

#0 CC   #1 CCCC

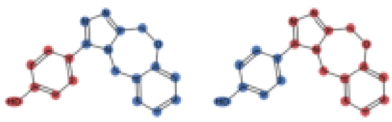


A1BKT · Q7ZWK8

4-[(4R)-5H,11H-[1,2,4]triazolo[3,4-c][1,4]benzoxazep…

```
family    OTOP1/PKD2L1 — sour taste channel
organism  Danio rerio
PDB       9MFM
fold 2   sites 6   pocket 58.7 res
flavour   (none) [llm_abstain]
SMILES    Oc1ccc(-c2nnc3n2Cc2ccccc2OC3)cc1
```

#0 Oc1ccccc1   #1 c1ccc2c(c1)Cn1cn… ●

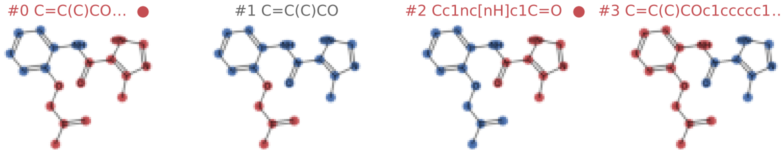


A1BKU · Q7ZWK8

4-methyl-N-{2-[(2-methylprop-2-en-1-yl)oxy]phenyl}-1…

```
family    OTOP1/PKD2L1 — sour taste channel
organism  Danio rerio
PDB       9MFF
fold 2   sites 4   pocket 55.0 res
flavour   (none) [llm_abstain]
SMILES    C=C(C)COc1ccccc1NC(=O)c1[nH]cnc1C
```

#0 C=C(C)CO… ●   #1 C=C(C)CO   #2 Cc1nc[nH]c1C=O ●   #3 C=C(C)COc1ccccc1… ●



Catalogue

records 117–120 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

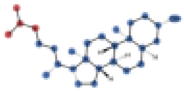
QNJ · Q7ZWK8

(3beta,5beta,14beta,17alpha)-cholestan-3-ol

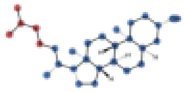
```
family    OTOP1/PKD2L1 — sour taste channel
organism  Danio rerio
PDB       9MFL
fold 2   sites 4   pocket 40.5 res
flavour   (none) [llm_abstain]
SMILES    CC(C)CCC[C@@H](C)[C@H]1CC[C@H]2[C@@H]3CC…
```

#0 CCC

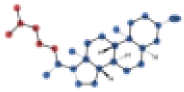
#1 CC(C)C



#2 CCC(C)C



#3 CCCC(C)C



A1ETA · noUP:9VQV

~{N}-(4-ethylphenyl)-2-[(4-ethyl-5-pyridin-3-yl-1,2,...

```
family    Odorant receptor (insect OR/Orco/IR)
organism  Drosophila bipectinata
PDB       9VQV
fold 0   sites 3   pocket 87.7 res
flavour   (none) [llm_abstain]
SMILES    CCc1ccc(NC(=O)CSc2nnc(-c3cccnc3)n2CC)cc1
```

#0 CCc1ccc(N)cc1...

#1 c1ccncc1

#2 CCc1ccc(N)cc1

#3 CCn1c(SCC=O)nnc1... ●

A1ETA · Q9VNB5;Q9VT92

~{N}-(4-ethylphenyl)-2-[(4-ethyl-5-pyridin-3-yl-1,2,...

```
family    Odorant receptor (insect OR/Orco/IR)
organism  Drosophila melanogaster
PDB       9VQR
fold 0   sites 3   pocket 87.0 res
flavour   (none) [llm_abstain]
SMILES    CCc1ccc(NC(=O)CSc2nnc(-c3cccnc3)n2CC)cc1
```

#0 CCc1ccc(N)cc1...

#1 c1ccncc1

#2 CCc1ccc(N)cc1

#3 CCn1c(SCC=O)nnc1... ●

A1ETW · noUP:9VQU

[(~{Z})-icos-11-enyl] ethanoate

```
family    Odorant receptor (insect OR/Orco/IR)
organism  Drosophila bipectinata
PDB       9VQU
fold 3   sites 1   pocket 116.0 res
flavour   (none) [llm_abstain]
SMILES    CCCCCCCC/C=C\CCCCCCCCCCOC(C)=O
```

#0 CCCCCCCCC CC=O

#1 CCCCCCCCC...

#2 C=C\CCCCCCCC...

#3 C=C\CCCCCCCC...

Catalogue

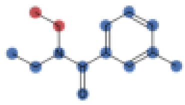
records 121–124 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

DE3 · noUP:7LIG

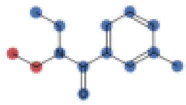
N,N-diethyl-3-methylbenzamide

```
family    Odorant receptor (insect OR/Orco/IR)
organism  Machilis hrabei
PDB       7LIG
fold 4   sites 4   pocket 68.0 res
flavour   (none) [llm_abstain]
SMILES    CCN(CC)C(=O)c1cccc(C)c1
```

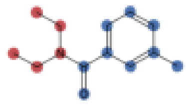
#0 CC



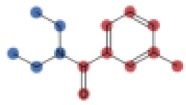
#1 CC



#2 CCNCC



#3 Cc1cccc(C=O)c1

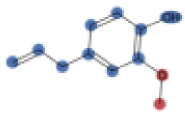


EOL · noUP:7LID

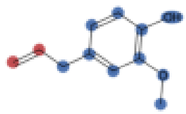
2-methoxy-4-(prop-2-en-1-yl)phenol

```
family    Odorant receptor (insect OR/Orco/IR)
organism  Machilis hrabei
PDB       7LID
fold 4   sites 4   pocket 63.0 res
flavour   spicy;sweet;woody [measured]
SMILES    C=CCc1ccc(O)c(OC)c1
```

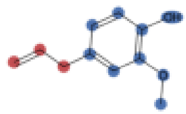
#0 CO



#1 C=C



#2 C=CC

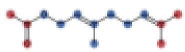


SOU · A0A1S6J137;A0A1S6J146

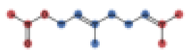
[(2E)-3,7-dimethylocta-2,6-dienyl] ethanoate

```
family    Odorant receptor (insect OR/Orco/IR)
organism  Acyrthosiphon pisum
PDB       8Z9A
fold 4   sites 1   pocket 81.0 res
flavour   bitter;fatty;floral;green [measured]
SMILES    CC(=O)OC/C=C(\C)CCC=C(C)C
```

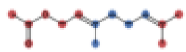
#0 CC=O CCC



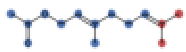
#1 CC(=O)O CCC



#2 CCOC(C)=O CCC



#3 CCC



VA · noUP:9VQT

(Z)-OCTADEC-11-ENYL ACETATE

```
family    Odorant receptor (insect OR/Orco/IR)
organism  Drosophila bipectinata
PDB       9VQT
fold 0   sites 1   pocket 111.0 res
flavour   (none) [llm_abstain]
SMILES    CCCCCC/C=C\CCCCCCCCCCOC(C)=O
```

#0 CCCCCCC CC=O

#1 CCCCCCC CC(=O)O

#2 C=C\CCCCCC CC=O

#3 C=C\CCCCCC...

Catalogue

records 125–128 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

VA · Q9VNB5;Q9VT92

(Z)-OCTADEC-11-ENYL ACETATE

family Odorant receptor (insect OR/Orco/IR)
organism Drosophila melanogaster
PDB 9VQP
fold 0 sites 1 pocket 106.0 res
flavour (none) [llm_abstain]
SMILES CCCCCC/C=C\CCCCCCCCCCOC(C)=O

#0 CCCCCCC CC=O

#1 CCCCCCC CC(=O)O

#2 C=C\CCCCCC CC=O

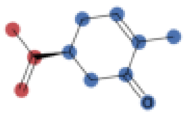
#3 C=C\CCCCCC...

OWU · P59768;P62873;P63092

(5S)-2-methyl-5-(prop-1-en-2-yl)cyclohex-2-en-1-one

family Olfactory receptor (vertebrate GPCR)
organism
PDB 8UY0
fold 4 sites 1 pocket 63.0 res
flavour roasted;spicy [measured]
SMILES C=C(C)[C@H]1CC=C(C)C(=O)C1

#0 C=CC

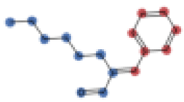


A1EJG · P38405;P59768;P62873

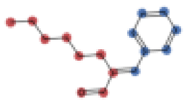
(2~{E})-2-(phenylmethylidene)octanal

family Olfactory receptor (vertebrate GPCR)
organism
PDB 9LDW
fold 0 sites 1 pocket 83.0 res
flavour fatty;floral;green;herbal [measured]
SMILES CCCCCC/C(C=O)=C\c1ccccc1

#0 Cc1ccccc1



#1 CCCCCCCC=O

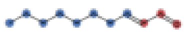


A1EJH · P38405;P59768;P62873

(~{E})-dec-2-enal

family Olfactory receptor (vertebrate GPCR)
organism
PDB 9LE1 9LE2
fold 0 sites 2 pocket 68.0 res
flavour citrus;earthy;fatty;floral;gre… [measured]
SMILES CCCCCCC/C=C/C=O

#0 CC=O



Catalogue

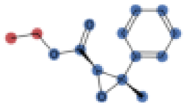
records 129–132 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

A1EJQ · P59768;P62873

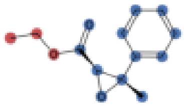
ethyl…

```
family    Olfactory receptor (vertebrate GPCR)
organism
PDB       9LDZ
fold 0   sites 1   pocket 68.0 res
flavour   fatty;floral;fruity;sweet [measured]
SMILES    CCOC(=O)[C@@H]1O[C@@]1(C)c1ccccc1
```

#0 CC



#1 CCO

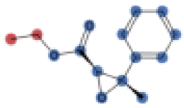


A1EJQ · P38405;P59768;P62873

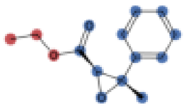
ethyl…

```
family    Olfactory receptor (vertebrate GPCR)
organism
PDB       9LDV
fold 0   sites 1   pocket 70.0 res
flavour   fatty;floral;fruity;sweet [measured]
SMILES    CCOC(=O)[C@@H]1O[C@@]1(C)c1ccccc1
```

#0 CC



#1 CCO



A1EKK · A2RT31;P62873

methyl (9R,10E,12E)-9-methoxyoctadeca-10,12-dienoate

```
family    Olfactory receptor (vertebrate GPCR)
organism  Mus musculus
PDB       9LKB
fold 3   sites 1   pocket 69.0 res
flavour   (none) [llm_abstain]
SMILES    CCCCC/C=C/C=C/[C@@H](CCCCCCCC(=O)OC)OC
```

#0 CCCCCC CO CO

#1 ? CO CO

#2 CO CO

#3 CCCCCC CO

A1EKK · A2RT31

methyl (9R,10E,12E)-9-methoxyoctadeca-10,12-dienoate

```
family    Olfactory receptor (vertebrate GPCR)
organism  Mus musculus
PDB       9LKD
fold 3   sites 1   pocket 65.0 res
flavour   (none) [llm_abstain]
SMILES    CCCCC/C=C/C=C/[C@@H](CCCCCCCC(=O)OC)OC
```

#0 CCCCCC CO CO

#1 ? CO CO

#2 CO CO

#3 CCCCCC CO

Catalogue

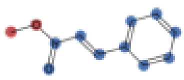
records 133–136 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

A1MBB · P59768;P62873;Q5FWY2;Q7TRW7

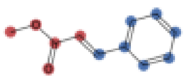
Methyl cinnamate

```
family    Olfactory receptor (vertebrate GPCR)
organism  Mus musculus
PDB       9W45
fold 4   sites 1   pocket 56.0 res
flavour   fruity;sweet;woody [measured]
SMILES    COC(=O)/C=C/c1ccccc1
```

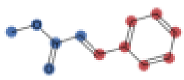
#0 CO



#1 CC(=O)OC



#2 Cc1ccccc1

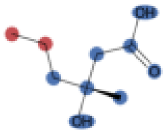


A1MBN · P59768;P62873

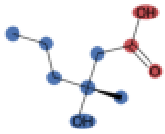
(3~{R})-3-methyl-3-oxidanyl-hexanoic acid

```
family    Olfactory receptor (vertebrate GPCR)
organism  Homo sapiens
PDB       9WG4
fold 0   sites 1   pocket 64.0 res
flavour   (none) [llm_abstain]
SMILES    CCC[C@@](C)(O)CC(=O)O
```

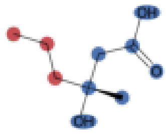
#0 CC



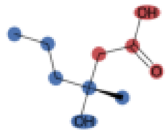
#1 O=CO



#2 CCC



#3 CC(=O)O

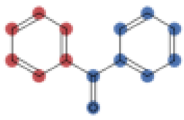


BZQ · P38405;P59768;P62873

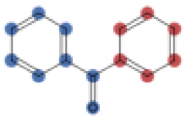
DIPHENYLMETHANONE

```
family    Olfactory receptor (vertebrate GPCR)
organism
PDB       9LDX
fold 0   sites 1   pocket 59.0 res
flavour   floral;fruity;medicinal;woody [measured]
SMILES    O=C(c1ccccc1)c1ccccc1
```

#0 c1ccccc1



#1 c1ccccc1



OCA · P59768;P62873;P63092

OCTANOIC ACID (CAPRYLIC ACID)

```
family    Olfactory receptor (vertebrate GPCR)
organism  Homo sapiens
PDB       8HTI
fold 4   sites 1   pocket 70.0 res
flavour   dairy;fatty;green [measured]
SMILES    CCCCCCCC(=O)O
```

#0 CC

#1 O=CO

#2 CCC

#3 CC(=O)O

Catalogue

records 137–140 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

OYA · P38405;P59768;P62873

OCTANAL

```
family    Olfactory receptor (vertebrate GPCR)
organism
PDB       9LE0
fold 0   sites 1   pocket 72.0 res
flavour   citrus;fatty;green [measured]
SMILES    CCCCCCCC=O
```

#0 C=O

#1 CC

#2 CC=O

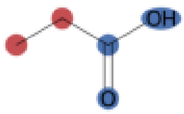
#3 CCC

PPI · P59768;P62873;P63092;Q9H255

PROPANOIC ACID

```
family    Olfactory receptor (vertebrate GPCR)
organism  Homo sapiens
PDB       8F76
fold 2   sites 1   pocket 51.0 res
flavour   dairy;fatty;sour;spicy [measured]
SMILES    CCC(=O)O
```

#0 CC

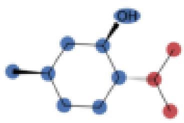


XUQ · P59768;P62873;P63092

(1R,2S,5R)-5-methyl-2-(propan-2-yl)cyclohexan-1-ol

```
family    Olfactory receptor (vertebrate GPCR)
organism
PDB       8UXY
fold 4   sites 1   pocket 58.0 res
flavour   minty;roasted;woody [measured]
SMILES    CC(C)[C@@H]1CC[C@@H](C)C[C@H]1O
```

#0 CCC

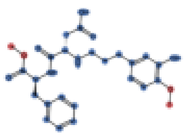


A1CD7 · Q8TE23;Q925D8

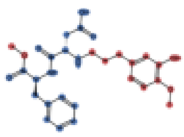
advantame

```
family    T1R — sweet/umami receptor
organism  Mus musculus
PDB       9OQ2 9OQ3
fold 2   sites 2   pocket 85.5 res
flavour   sweet [llm]
SMILES    COC(=O)[C@H](Cc1ccccc1)NC(=O)[C@H](CC(=O…
```

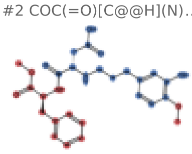
#0 CO CO



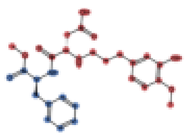
#1 CO…



#2 COC(=O)[C@@H](N)…



#3 CO…



Catalogue

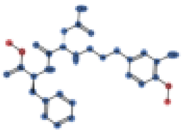
records 141–144 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

A1CD7 · Q7RTX0;Q8TE23

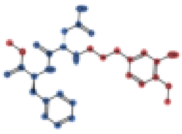
advantame

```
family    T1R — sweet/umami receptor
organism  Homo sapiens
PDB       9OQ4 9OQ5 9OQ6
fold 2   sites 3   pocket 86.0 res
flavour   sweet [llm]
SMILES    COC(=O)[C@H](Cc1ccccc1)NC(=O)[C@H](CC(=O…
```

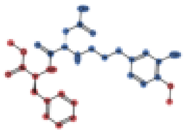
#0 CO CO



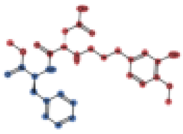
#1 CO...



#2 COC(=O)[C@@H](N)...



#3 CO...

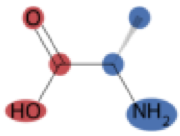


ALA · A0A173M094;A0A173M0G2

ALANINE

```
family    T1R — sweet/umami receptor
organism  Oryzias latipes
PDB       5X2N
fold 4   sites 4   pocket 51.2 res
flavour   odorless [measured]
SMILES    C[C@H](N)C(=O)O
```

#0 O=CO



ARG · A0A173M094;A0A173M0G2

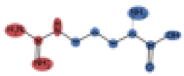
ARGININE

```
family    T1R — sweet/umami receptor
organism  Oryzias latipes
PDB       5X2O
fold 4   sites 4   pocket 64.8 res
flavour   odorless [measured]
SMILES    NC(=[NH2+])NCCC[C@H](N)C(=O)O
```

#0 O=CO

#1 NC=[NH2+] ●

#2 NC(N)=[NH2+] ●



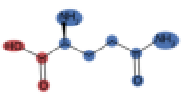
#3 NCC(=O)O

GLN · A0A173M094;A0A173M0G2

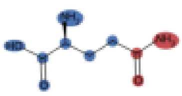
GLUTAMINE

```
family    T1R — sweet/umami receptor
organism  Oryzias latipes
PDB       5X2M
fold 4   sites 4   pocket 63.5 res
flavour   dairy;fatty;odorless;roasted [measured]
SMILES    NC(=O)CC[C@H](N)C(=O)O
```

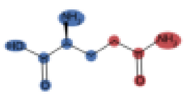
#0 O=CO



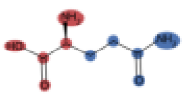
#1 NC=O



#2 CC(N)=O



#3 NCC(=O)O



Catalogue

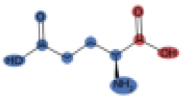
records 145–148 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

GLU · A0A173M094;A0A173M0G2

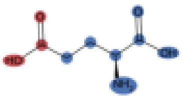
GLUTAMIC ACID

```
family    T1R — sweet/umami receptor
organism  Oryzias latipes
PDB       5X2P
fold 4   sites 4   pocket 63.5 res
flavour   dairy;odorless;roasted [measured]
SMILES    N[C@@H](CCC(=O)O)C(=O)O
```

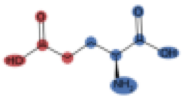
#0 O=CO



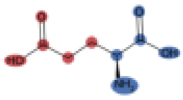
#1 O=CO



#2 CC(=O)O



#3 CCC(=O)O

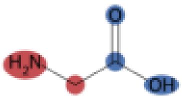


GLY · A0A173M094;A0A173M0G2

GLYCINE

```
family    T1R — sweet/umami receptor
organism  Oryzias latipes
PDB       5X2Q
fold 4   sites 4   pocket 51.0 res
flavour   fatty;odorless;sweet [measured]
SMILES    NCC(=O)O
```

#0 CN

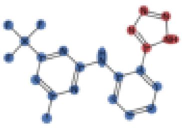


A1AEI · Q9NYV8

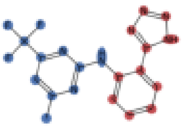
4-methyl-N-[(2M)-2-(1H-tetrazol-5-yl)phenyl]-6-(trif...

```
family    T2R — bitter taste receptor
organism  Homo sapiens
PDB       9IIW 9IJA
fold 3   sites 3   pocket 67.7 res
flavour   (none) [llm_abstain]
SMILES    Cc1cc(C(F)(F)F)nc(Nc2ccccc2-c2nnn[nH]2)n…
```

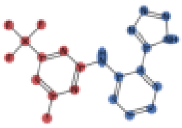
#0 c1nnn[nH]1



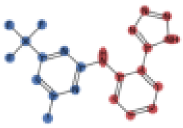
#1 c1ccc(-c2nnn[nH]...



#2 Cc1cc(C(F)(F)F)n... ●



#3 Nc1ccccc1-c1nnn[...

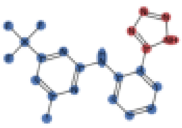


A1AEI · A8MTJ3;P59768;P62873;Q9NYV8

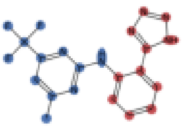
4-methyl-N-[(2M)-2-(1H-tetrazol-5-yl)phenyl]-6-(trif...

```
family    T2R — bitter taste receptor
organism  Homo sapiens
PDB       8VY9
fold 3   sites 1   pocket 76.0 res
flavour   (none) [llm_abstain]
SMILES    Cc1cc(C(F)(F)F)nc(Nc2ccccc2-c2nnn[nH]2)n…
```

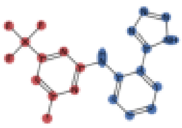
#0 c1nnn[nH]1



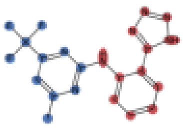
#1 c1ccc(-c2nnn[nH]...



#2 Cc1cc(C(F)(F)F)n... ●



#3 Nc1ccccc1-c1nnn[...



Catalogue

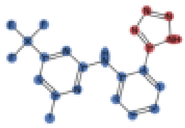
records 149–152 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

A1AEI · P59768;P62873;P63096;Q9NYV8

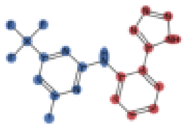
4-methyl-N-[(2M)-2-(1H-tetrazol-5-yl)phenyl]-6-(trif…

```
family    T2R — bitter taste receptor
organism  Homo sapiens
PDB       8VY7 9IJ9
fold 3   sites 2   pocket 76.5 res
flavour   (none) [llm_abstain]
SMILES    Cc1cc(C(F)(F)F)nc(Nc2ccccc2-c2nnn[nH]2)n…
```

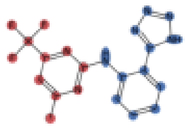
#0 c1nnn[nH]1



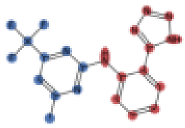
#1 c1ccc(-c2nnn[nH]…



#2 Cc1cc(C(F)(F)F)n… ●



#3 Nc1ccccc1-c1nnn[…

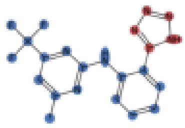


A1AEI · P59768;P62873;Q9NYV8

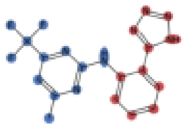
4-methyl-N-[(2M)-2-(1H-tetrazol-5-yl)phenyl]-6-(trif…

```
family    T2R — bitter taste receptor
organism  Homo sapiens
PDB       9IIX
fold 3   sites 2   pocket 79.5 res
flavour   (none) [llm_abstain]
SMILES    Cc1cc(C(F)(F)F)nc(Nc2ccccc2-c2nnn[nH]2)n…
```

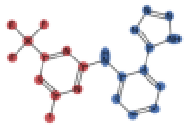
#0 c1nnn[nH]1



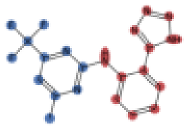
#1 c1ccc(-c2nnn[nH]…



#2 Cc1cc(C(F)(F)F)n… ●



#3 Nc1ccccc1-c1nnn[…

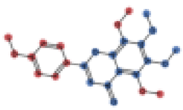


A1EUI · P59540;P59768;P62873;P63096

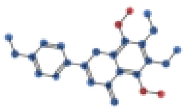
5,6,7,8-tetramethoxy-2-(4-methoxyphenyl)chromen-4-on…

```
family    T2R — bitter taste receptor
organism  Homo sapiens
PDB       9W0U
fold 3   sites 1   pocket 66.0 res
flavour   bitter [measured]
SMILES    COc1ccc(-c2cc(=O)c3c(OC)c(OC)c(OC)c(OC)c…
```

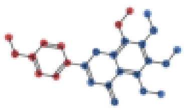
#0 COc1ccccc1 CO CO



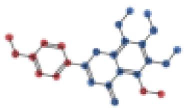
#1 CO CO



#2 COc1ccccc1 CO



#3 COc1ccccc1 CO

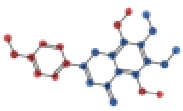


A1EUI · P59768;P62873;P63096;Q9NYV8

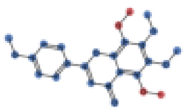
5,6,7,8-tetramethoxy-2-(4-methoxyphenyl)chromen-4-on…

```
family    T2R — bitter taste receptor
organism  Escherichia coli
PDB       9W0Q
fold 3   sites 1   pocket 70.0 res
flavour   bitter [measured]
SMILES    COc1ccc(-c2cc(=O)c3c(OC)c(OC)c(OC)c(OC)c…
```

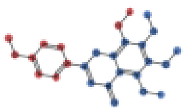
#0 COc1ccccc1 CO CO



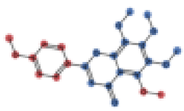
#1 CO CO



#2 COc1ccccc1 CO



#3 COc1ccccc1 CO



Catalogue

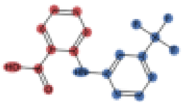
records 153–156 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

FLF · P59768;P62873;P63092;Q59310;Q9…

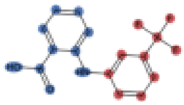
2-[[3-(TRIFLUOROMETHYL)PHENYL]AMINO] BENZOIC ACID

```
family    T2R — bitter taste receptor
organism  Homo sapiens
PDB       8XQR
fold 3   sites 1   pocket 73.0 res
flavour   bitter [measured]
SMILES    O=C(O)c1ccccc1Nc1cccc(C(F)(F)F)c1
```

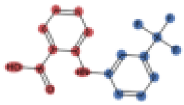
#0 O=C(O)c1ccccc1



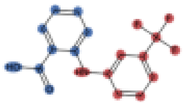
#1 FC(F)(F)c1ccccc1



#2 Nc1ccccc1C(=O)O



#3 Nc1cccc(C(F)(F)F…

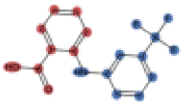


FLF · A8MTJ3;P59768;P62873;Q9NYV8

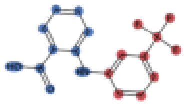
2-[[3-(TRIFLUOROMETHYL)PHENYL]AMINO] BENZOIC ACID

```
family    T2R — bitter taste receptor
organism  Homo sapiens
PDB       8RQL
fold 3   sites 2   pocket 70.5 res
flavour   bitter [measured]
SMILES    O=C(O)c1ccccc1Nc1cccc(C(F)(F)F)c1
```

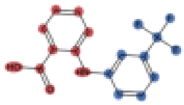
#0 O=C(O)c1ccccc1



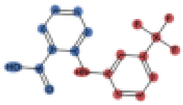
#1 FC(F)(F)c1ccccc1



#2 Nc1ccccc1C(=O)O



#3 Nc1cccc(C(F)(F)F…

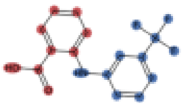


FLF · P59768;P62873;P63096;Q59310;Q9…

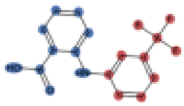
2-[[3-(TRIFLUOROMETHYL)PHENYL]AMINO] BENZOIC ACID

```
family    T2R — bitter taste receptor
organism  Homo sapiens
PDB       8XQS
fold 3   sites 1   pocket 70.0 res
flavour   bitter [measured]
SMILES    O=C(O)c1ccccc1Nc1cccc(C(F)(F)F)c1
```

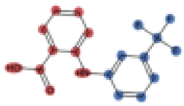
#0 O=C(O)c1ccccc1



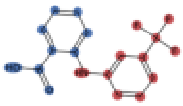
#1 FC(F)(F)c1ccccc1



#2 Nc1ccccc1C(=O)O



#3 Nc1cccc(C(F)(F)F…

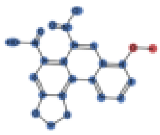


GOQ · P59768;P62873;P63096;Q59310;Q9…

8-methoxy-6-nitro-naphtho[1,2-e][1,3]benzodioxole-5-…

```
family    T2R — bitter taste receptor
organism  Homo sapiens
PDB       8XQN 8XQO
fold 3   sites 2   pocket 75.5 res
flavour   bitter [measured]
SMILES    COc1cccc2c1cc([N+](=O)[O-])c1c(C(=O)O)cc…
```

#0 CO



Catalogue

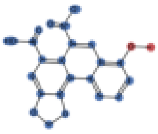
records 157–160 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

GOQ · A8MTJ3;P59537;P59768;P62873

8-methoxy-6-nitro-naphtho[1,2-e][1,3]benzodioxole-5-…

```
family    T2R — bitter taste receptor
organism  Homo sapiens
PDB       9OXB
fold 3   sites 1   pocket 73.0 res
flavour   bitter [measured]
SMILES    COc1cccc2c1cc([N+](=O)[O-])c1c(C(=O)O)cc…
```

#0 CO

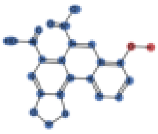


GOQ · P0ABE7;P59537;P59768;P62873;P6…

8-methoxy-6-nitro-naphtho[1,2-e][1,3]benzodioxole-5-…

```
family    T2R — bitter taste receptor
organism  Homo sapiens
PDB       9OXA
fold 3   sites 1   pocket 75.0 res
flavour   bitter [measured]
SMILES    COc1cccc2c1cc([N+](=O)[O-])c1c(C(=O)O)cc…
```

#0 CO

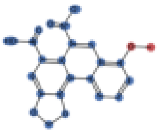


GOQ · A8MTJ3;P59768;P62873;Q59310;Q9…

8-methoxy-6-nitro-naphtho[1,2-e][1,3]benzodioxole-5-…

```
family    T2R — bitter taste receptor
organism  Homo sapiens
PDB       8XQP
fold 3   sites 1   pocket 73.0 res
flavour   bitter [measured]
SMILES    COc1cccc2c1cc([N+](=O)[O-])c1c(C(=O)O)cc…
```

#0 CO

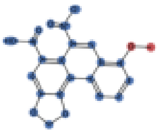


GOQ · P59768;P62873;P63092;Q59310;Q9…

8-methoxy-6-nitro-naphtho[1,2-e][1,3]benzodioxole-5-…

```
family    T2R — bitter taste receptor
organism  Homo sapiens
PDB       8XQL
fold 3   sites 2   pocket 74.0 res
flavour   bitter [measured]
SMILES    COc1cccc2c1cc([N+](=O)[O-])c1c(C(=O)O)cc…
```

#0 CO



Catalogue

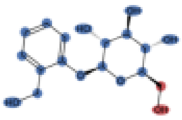
records 161–164 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

SA0 · P04899;P59768;P62873;Q9NYV7

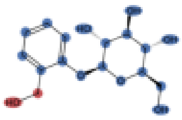
2-(hydroxymethyl)phenyl beta-D-glucopyranoside

```
family    T2R — bitter taste receptor
organism  Homo sapiens
PDB       9K6L
fold 2   sites 1   pocket 70.0 res
flavour   bitter [measured]
SMILES    OCc1ccccc1O[C@@H]1O[C@H](CO)[C@@H](O)[C@…
```

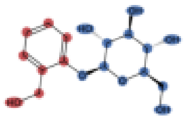
#0 CO



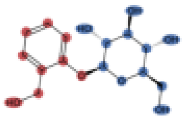
#1 CO



#2 OCc1ccccc1



#3 OCc1ccccc1O

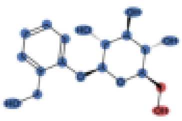


SA0 · P59768;P62873;P63092;Q9NYV7

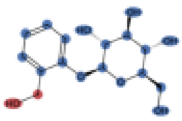
2-(hydroxymethyl)phenyl beta-D-glucopyranoside

```
family    T2R — bitter taste receptor
organism  Homo sapiens
PDB       9KPD
fold 2   sites 1   pocket 68.0 res
flavour   bitter [measured]
SMILES    OCc1ccccc1O[C@@H]1O[C@H](CO)[C@@H](O)[C@…
```

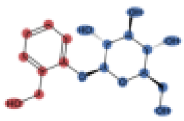
#0 CO



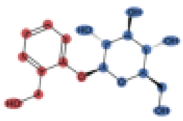
#1 CO



#2 OCc1ccccc1



#3 OCc1ccccc1O

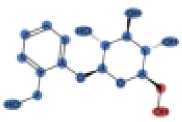


SA0 · P59768;P62873;P63096;Q9NYV7

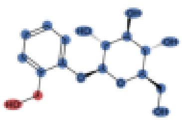
2-(hydroxymethyl)phenyl beta-D-glucopyranoside

```
family    T2R — bitter taste receptor
organism  Homo sapiens
PDB       9KPE 9KPF
fold 2   sites 2   pocket 68.0 res
flavour   bitter [measured]
SMILES    OCc1ccccc1O[C@@H]1O[C@H](CO)[C@@H](O)[C@…
```

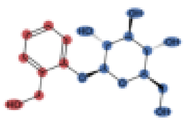
#0 CO



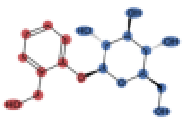
#1 CO



#2 OCc1ccccc1



#3 OCc1ccccc1O

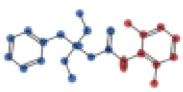


WK3 · P59540;P59768;P62873;P63096

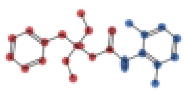
N-benzyl-2-(2,6-dimethylanilino)-N,N-diethyl-2-oxoet…

```
family    T2R — bitter taste receptor
organism  Homo sapiens
PDB       9W0T
fold 3   sites 1   pocket 57.0 res
flavour   (none) [llm_abstain]
SMILES    CC[N+](CC)(CC(=O)Nc1c(C)cccc1C)Cc1ccccc1
```

#0 Cc1cccc(C)c1N



#1 CC[N+](CC)(CC=O)…



Catalogue

records 165–168 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

XC9 · P0ABE7;P59768;P62873;P63096;Q9…

1-[6-[azanylidene-[[azanylidene-[[(4-chlorophenyl)am…

```
family    T2R — bitter taste receptor
organism  Escherichia coli
PDB       9W0P
fold 4   sites 1   pocket 69.0 res
flavour   bitter [measured]
SMILES    [H]/N=C(/NCCCCCCN/C(=N\[H])N/C(=N\[H])Nc…
```

#0 Clc1ccccc1

#1 Clc1ccccc1

#2 Nc1ccc(Cl)cc1

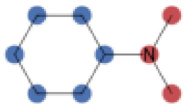
#3 Nc1ccc(Cl)cc1

8IA · P59768;P62873;P63092;Q5QD08

~{N},~{N}-dimethylcyclohexanamine

```
family    TAAR — olfactory amine receptor…
organism  Mus musculus
PDB       8PM2
fold 3   sites 1   pocket 54.0 res
flavour   (none) [llm_abstain]
SMILES    CN(C)C1CCCCC1
```

#0 CNC

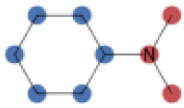


8IA · P59768;P62873;Q5QD04

~{N},~{N}-dimethylcyclohexanamine

```
family    TAAR — olfactory amine receptor…
organism  Mus musculus
PDB       8ITF
fold 3   sites 1   pocket 54.0 res
flavour   (none) [llm_abstain]
SMILES    CN(C)C1CCCCC1
```

#0 CNC



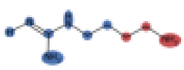
AG2 · P59768;P62873;Q5QNQ1

AGMATINE

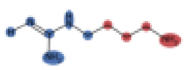
```
family    TAAR — olfactory amine receptor…
organism  Danio rerio
PDB       9M0X
fold 4   sites 2   pocket 59.0 res
flavour   bitter [mined]
SMILES    [H]/N=C(\N)NCCCCN
```

#0 CN

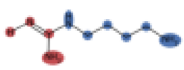
#1 CCN



#2 CCCN



#3 [H]/N=C\N



Catalogue

records 169–172 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

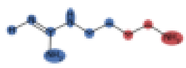
AG2 · P59768;P62873;P63092;Q5QNP8

AGMATINE

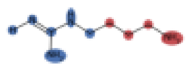
```
family    TAAR — olfactory amine receptor…
organism  Danio rerio
PDB       9M40
fold 4   sites 1   pocket 64.0 res
flavour   bitter [mined]
SMILES    [H]/N=C(\N)NCCCCN
```

#0 CN

#1 CCN



#2 CCCN



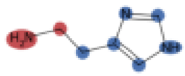
#3 [H]/N=C\N

HSM · P59768;P62873;Q5QNP9

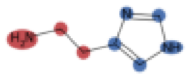
HISTAMINE

```
family    TAAR — olfactory amine receptor…
organism  Danio rerio
PDB       9M3S
fold 4   sites 1   pocket 62.0 res
flavour   bitter [mined]
SMILES    NCCc1c[nH]cn1
```

#0 CN



#1 CCN

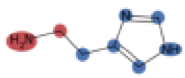


HSM · P59768;P62873;Q5QNQ1

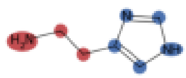
HISTAMINE

```
family    TAAR — olfactory amine receptor…
organism  Danio rerio
PDB       9M1P
fold 4   sites 1   pocket 62.0 res
flavour   bitter [mined]
SMILES    NCCc1c[nH]cn1
```

#0 CN



#1 CCN



N2P · P59768;P62873;Q5QNP2

PENTANE-1,5-DIAMINE

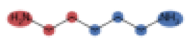
```
family    TAAR — olfactory amine receptor…
organism  Danio rerio
PDB       9M2S
fold 4   sites 1   pocket 59.0 res
flavour   bitter [mined]
SMILES    NCCCCCN
```

#0 CN

#1 CN

#2 CCN

#3 CCN



Catalogue

records 173–176 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

N2P · P59768;P62873;Q5QNQ1

PENTANE-1,5-DIAMINE

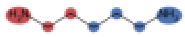
```
family    TAAR — olfactory amine receptor…
organism  Danio rerio
PDB       9M0Z
fold 4   sites 1   pocket 60.0 res
flavour   bitter [mined]
SMILES    NCCCCCN
```

#0 CN

#1 CN

#2 CCN

#3 CCN

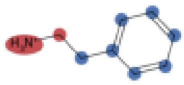


PEA · P59768;P62873;P63092;P63096;Q5…

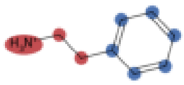
2-PHENYLETHYLAMINE

```
family    TAAR — olfactory amine receptor…
organism  Mus musculus
PDB       8IW7
fold 4   sites 1   pocket 60.0 res
flavour   meaty;medicinal [measured]
SMILES    [NH3+]CCc1ccccc1
```

#0 C[NH3+] ●



#1 CC[NH3+]

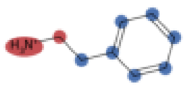


PEA · Q5QD04

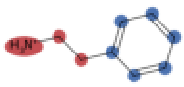
2-PHENYLETHYLAMINE

```
family    TAAR — olfactory amine receptor…
organism  Mus musculus
PDB       8IWM
fold 4   sites 1   pocket 60.0 res
flavour   meaty;medicinal [measured]
SMILES    [NH3+]CCc1ccccc1
```

#0 C[NH3+] ●



#1 CC[NH3+]

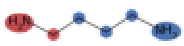


PUT · P59768;P62873;Q5QNQ1

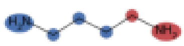
1,4-DIAMINOBUTANE

```
family    TAAR — olfactory amine receptor…
organism  Danio rerio
PDB       9M2O
fold 4   sites 1   pocket 61.0 res
flavour   (none) [llm_abstain]
SMILES    NCCCCN
```

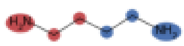
#0 CN



#1 CN



#2 CCN



Catalogue

records 177–180 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

SPD · Q5QD04

SPERMIDINE

```
family    TAAR — olfactory amine receptor…
organism  Mus musculus
PDB       8IWE
fold 4   sites 1   pocket 61.0 res
flavour   (none) [llm_abstain]
SMILES    NCCCCNCCCN
```

#0 CN CCCN

#1 CN

#2 CCCN

#3 CCCN

SPD · P59768;P62873;P63092;P63096;Q5…

SPERMIDINE

```
family    TAAR — olfactory amine receptor…
organism  Mus musculus
PDB       8IW4
fold 4   sites 1   pocket 61.0 res
flavour   (none) [llm_abstain]
SMILES    NCCCCNCCCN
```

#0 CN CCCN

#1 CN

#2 CCCN

#3 CCCN

SPD · P59768;P62873;Q5QNP2

SPERMIDINE

```
family    TAAR — olfactory amine receptor…
organism  Danio rerio
PDB       9M3Q
fold 4   sites 1   pocket 71.0 res
flavour   (none) [llm_abstain]
SMILES    NCCCCNCCCN
```

#0 CN CCCN

#1 CN

#2 CCCN

#3 CCCN

SPM · A0A590UJY2;A0A678XLR1;P59768;P…

SPERMINE

```
family    TAAR — olfactory amine receptor…
organism  Petromyzon marinus
PDB       9VMG
fold 2   sites 1   pocket 72.0 res
flavour   (none) [llm_abstain]
SMILES    NCCCNCCCCNCCCN
```

#0 CCCN CCCN

#1 CCCN NCCCN

#2 NCCCN CCCN

#3 CCCN

Catalogue

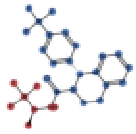
records 181–184 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

A1AIY · Q8R4D5

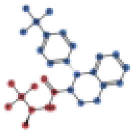
(8R)-8-[4-(trifluoromethyl)phenyl]-N-[(2S)-1,1,1-tri...

```
family    TRP taste/chemesthesis channel
organism  Mus musculus
PDB       9B6F
fold 4    sites 4    pocket 72.8 res
flavour   (none) [llm_abstain]
SMILES    C[C@H](NC(=O)N1CCc2cccnc2[C@H]1c1ccc(C(F…
```

#0 C[C@H](N)C(F)(F)... ●



#1 C[C@H](NC=O)C(F)... ●

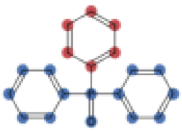


A1CG5 · S5UH55

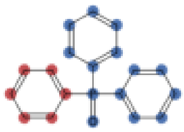
oxotri(phenyl)-lambda~5~-phosphane

```
family    TRP taste/chemesthesis channel
organism  Danio rerio
PDB       9P3T
fold 3    sites 4    pocket 77.0 res
flavour   (none) [llm_abstain]
SMILES    O=P(c1ccccc1)(c1ccccc1)c1ccccc1
```

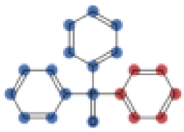
#0 c1ccccc1



#1 c1ccccc1



#2 c1ccccc1

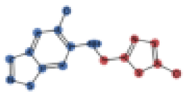


A1CGW · A0A5P9VSM8;S5UH55

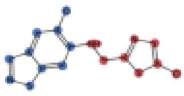
5-chloro-N-[(5-chloro-1,3-thiazol-2-yl)methyl]-1,2-b...

```
family    TRP taste/chemesthesis channel
organism  Danio rerio
PDB       9P3U 9P3V
fold 3    sites 8    pocket 71.5 res
flavour   (none) [llm_abstain]
SMILES    Clc1cnc(CNc2cc3sncc3cc2Cl)s1
```

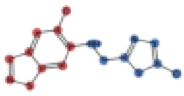
#0 Cc1ncc(Cl)s1 ●



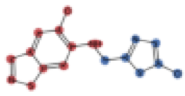
#1 NCc1ncc(Cl)s1 ●



#2 Clc1ccc2sncc2c1 ●



#3 Nc1cc2sncc2cc1Cl ●

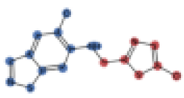


A1CGW · S5UH55

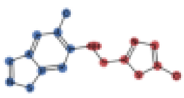
5-chloro-N-[(5-chloro-1,3-thiazol-2-yl)methyl]-1,2-b...

```
family    TRP taste/chemesthesis channel
organism  Danio rerio
PDB       9P3N 9P3O 9P3P
fold 3    sites 12    pocket 69.7 res
flavour   (none) [llm_abstain]
SMILES    Clc1cnc(CNc2cc3sncc3cc2Cl)s1
```

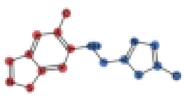
#0 Cc1ncc(Cl)s1 ●



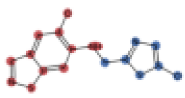
#1 NCc1ncc(Cl)s1 ●



#2 Clc1ccc2sncc2c1 ●



#3 Nc1cc2sncc2cc1Cl ●



Catalogue

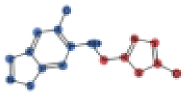
records 185–188 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

A1CGW · A0AB32TV16

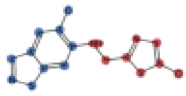
5-chloro-N-[(5-chloro-1,3-thiazol-2-yl)methyl]-1,2-b...

```
family    TRP taste/chemesthesis channel
organism  Danio rerio
PDB       9P3S
fold 3   sites 4   pocket 68.0 res
flavour   (none) [llm_abstain]
SMILES    Clc1cnc(CNc2cc3sncc3cc2Cl)s1
```

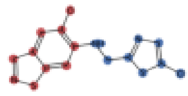
#0 Cc1ncc(Cl)s1 ●



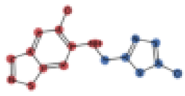
#1 NCc1ncc(Cl)s1 ●



#2 Clc1ccc2sncc2c1 ●



#3 Nc1cc2sncc2cc1Cl ●

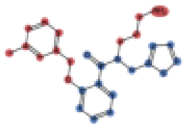


LQ7 · Q8R4D5

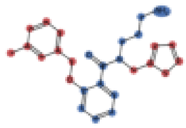
N-(3-aminopropyl)-2-[(3-methylphenyl)methoxy]-N-[(th...

```
family    TRP taste/chemesthesis channel
organism  Mus musculus
PDB       9B6G
fold 4   sites 4   pocket 87.0 res
flavour   (none) [llm_abstain]
SMILES    Cc1cccc(COc2ccccc2C(=O)N(CCCN)Cc2cccs2)c…
```

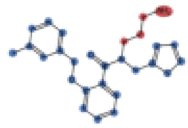
#0 Cc1cccc(CO)c1...



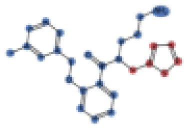
#1 Cc1cccc(CO)c1...



#2 CCCN



#3 Cc1cccs1

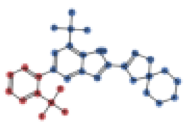


T14 · A0A5S8WF66

3-{7-(trifluoromethyl)-5-[2-(trifluoromethyl)phenyl]...

```
family    TRP taste/chemesthesis channel
organism  Parus major
PDB       6O72 9B6I
fold 4   sites 8   pocket 88.5 res
flavour   (none) [llm_abstain]
SMILES    FC(F)(F)c1ccccc1-c1cc(C(F)(F)F)c2[nH]c(C…
```

#0 FC(F)(F)c1ccccc1

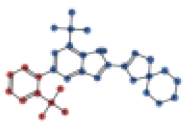


T14 · Q8R4D5

3-{7-(trifluoromethyl)-5-[2-(trifluoromethyl)phenyl]...

```
family    TRP taste/chemesthesis channel
organism  Mus musculus
PDB       9B6E 9B6H
fold 4   sites 8   pocket 89.8 res
flavour   (none) [llm_abstain]
SMILES    FC(F)(F)c1ccccc1-c1cc(C(F)(F)F)c2[nH]c(C…
```

#0 FC(F)(F)c1ccccc1



Catalogue

records 189–192 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

ULO · Q8R4D5

nonyl(oxo)di(propan-2-yl)-lambda~5~-phosphane

```
family    TRP taste/chemesthesis channel
organism  Mus musculus
PDB       8E4L 8E4M 9B6H
fold 4   sites 12   pocket 81.7 res
flavour   (none) [llm_abstain]
SMILES    CCCCCCCCCP(=O)(C(C)C)C(C)C
```

#0 CC

#1 CCC

#2 CCC

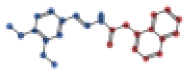
#3 CCC

YUS · S5UH55

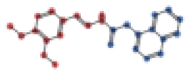
N'-[(E)-(3,4-dimethoxyphenyl)methylidene]-2-(naphtha…

```
family    TRP taste/chemesthesis channel
organism  Danio rerio
PDB       7MBV
fold 3   sites 4   pocket 62.0 res
flavour   (none) [llm_abstain]
SMILES    COc1ccc(/C=N/NC(=O)Cc2cccc3ccccc23)cc1OC
```

#0 O=CCc1cccc2ccccc…



#1 COc1ccc(/C=N/N)c…



0IG · O75762

~{N}-[4-[4-[(azanylidene-$l^{4}-azanylidene)amino]ph…

```
family    TRPA1 — irritant/pungency receptor…
organism  Homo sapiens
PDB       7OR0 7OR1
fold 1   sites 8   pocket 58.1 res
flavour   (none) [llm_abstain]
SMILES    Cn1c(=O)c2c(ncn2CC(=O)Nc2nc(-c3ccc(N=[N+…
```

#0 N=[N+]=Nc1ccccc1 ●

#1 Cn1c(=O)c2ncnc2n…

#2 N=[N+]=Nc1ccc(-c… ●

#3 Cn1c(=O)c2c(ncn2…

6OU · O75762

[(2~{R})-1-[2-azanylethoxy(oxidanyl)phosphoryl]oxy-3…

```
family    TRPA1 — irritant/pungency receptor…
organism  Homo sapiens
PDB       6PQO 6PQP 6PQQ
fold 1   sites 56   pocket 43.9 res
flavour   (none) [llm_abstain]
SMILES    CCCCCCCC/C=C\CCCCCCCC(=O)O[C@H](COC(=O)C…
```

#0 CCCCCCCCC…

#1 CCCCCCCCC…

#2 CCCCCCCCC…

#3 CCCCCCCCC…

Catalogue

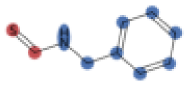
records 193–196 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

9BE · O75762

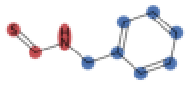
N-benzylthioformamide

```
family    TRPA1 — irritant/pungency receptor…
organism  Homo sapiens
PDB       6PQP
fold 1    sites 4    pocket 50.0 res
flavour   (none) [llm_abstain]
SMILES    S=CNCc1ccccc1
```

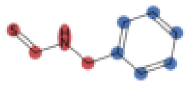
#0 C=S



#1 NC=S



#2 CNC=S



A1BNO · O75762

(2S)-2-{3-methyl-1-[(3-methyl-1,2,4-oxadiazol-5-yl)m…

```
family    TRPA1 — irritant/pungency receptor…
organism  Homo sapiens
PDB       9MOE
fold 1    sites 4    pocket 66.8 res
flavour   (none) [llm_abstain]
SMILES    Cc1noc(Cn2c(=O)c3c(ncn3[C@@H](C)C(=O)Nc3…
```

#0 Cc1noc(C)n1… ●

#1 Cc1noc(C)n1… ●

#2 Cc1noc(C)n1

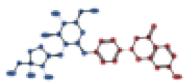
#3 FC(F)(F)c1ncccn1 ●

A1ETS · O75762

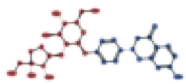
Liquiritin apioside

```
family    TRPA1 — irritant/pungency receptor…
organism  Homo sapiens
PDB       9VU1
fold 1    sites 4    pocket 55.2 res
flavour   (none) [llm_abstain]
SMILES    O=C1C[C@@H](c2ccc(O[C@@H]3O[C@H](CO)[C@@…
```

#0 O=C1C[C@@H](c2cc…



#1 OC[C@H]1O[C@@H](…

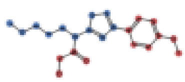


JT0 · O75762

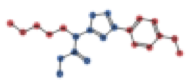
2-chloro-N-[4-(4-methoxyphenyl)-1,3-thiazol-2-yl]-N-…

```
family    TRPA1 — irritant/pungency receptor…
organism  Homo sapiens
PDB       6PQO
fold 1    sites 4    pocket 48.0 res
flavour   (none) [llm_abstain]
SMILES    COCCCN(C(=O)CCl)c1nc(-c2ccc(OC)cc2)cs1
```

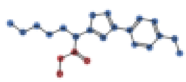
#0 O=CCCl…



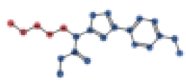
#1 CCCOC COc1ccccc1



#2 O=CCCl



#3 CCCOC



Catalogue

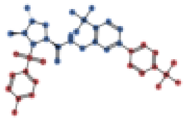
records 197–200 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

LXY · O75762

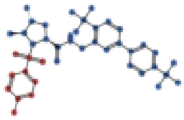
(4R,5S)-4-fluoro-1-[(4-fluorophenyl)sulfonyl]-5-meth…

```
family     TRPA1 — irritant/pungency receptor…
organism   Homo sapiens
PDB        6WJ5
fold 1    sites 4    pocket 90.8 res
flavour    (none) [llm_abstain]
SMILES     C[C@H]1[C@H](F)C[C@@H](C(=O)NCc2cc(-c3cn…
```

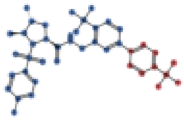
#0 O=S(=O)c1ccc(F)c… ●



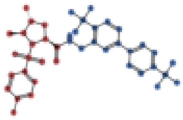
#1 O=S(=O)c1ccc(F)c…



#2 FC(F)(F)c1ncccn1 ●



#3 C[C@H]1[C@H](F)C… ●

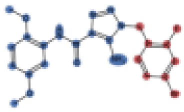


ULJ · O75762

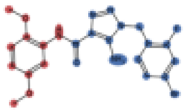
5-amino-1-[(4-bromo-2-fluorophenyl)methyl]-N-(2,5-di…

```
family     TRPA1 — irritant/pungency receptor…
organism   Homo sapiens
PDB        6X2J
fold 1    sites 4    pocket 81.2 res
flavour    (none) [llm_abstain]
SMILES     COc1ccc(OC)c(NC(=O)c2nnn(Cc3ccc(Br)cc3F)…
```

#0 Cc1ccc(Br)cc1F ●



#1 COc1ccc(OC)c(N)c…



#2 Nc1c(C=O)nnn1Cc1… ●

VKM · O75762

1-({3-[(3R,5R)-5-(4-fluorophenyl)oxolan-3-yl]-1,2,4-…

```
family     TRPA1 — irritant/pungency receptor…
organism   Homo sapiens
PDB        7JUP
fold 1    sites 4    pocket 75.0 res
flavour    (none) [llm_abstain]
SMILES     Cn1cnc2ncn(Cc3nc([C@@H]4CO[C@@H](c5ccc(F…
```

#0 Cn1cnc2ncnc(=O)c… ●

#1 Cc1nc([C@@H]2CO[… ●

KX7 · noUP:6NR3

Icilin

```
family     TRPM8 — cooling/menthol receptor…
organism   Ficedula albicollis
PDB        6NR3
fold 4    sites 4    pocket 72.0 res
flavour    minty [llm]
SMILES     O=C1NC(c2cccc([N+](=O)[O-])c2)=CCN1c1ccc…
```

#0 Oc1ccccc1

Catalogue

records 201–204 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

KX7 · Q8R4D5

Icilin

family TRPM8 – cooling/menthol receptor…
organism Mus musculus
PDB 7WRC 7WRD 7WRE 7WRF 9VJR 9VNC
fold 4 sites 24 pocket 69.8 res
flavour minty [llm]
SMILES O=C1NC(c2cccc([N+](=O)[O-])c2)=CCN1c1ccc…

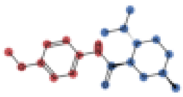
#0 Oc1ccccc1

KXS · noUP:6NR2

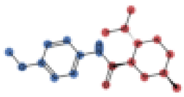
(1R,2S,5R)-N-(4-methoxyphenyl)-5-methyl-2-(propan-2-…

family TRPM8 – cooling/menthol receptor…
organism Ficedula albicollis
PDB 6NR2
fold 3 sites 4 pocket 79.0 res
flavour minty [llm]
SMILES COc1ccc(NC(=O)[C@@H]2C[C@H](C)CC[C@H]2C(…

#0 COc1ccc(N)cc1



#1 CC(C)[C@@H]1CC[C…

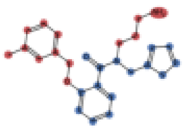


LQ7 · A0A5S8WF66

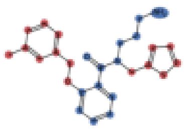
N-(3-aminopropyl)-2-[(3-methylphenyl)methoxy]-N-[(th…

family TRPM8 – cooling/menthol receptor…
organism Parus major
PDB 6O6R
fold 4 sites 4 pocket 82.0 res
flavour (none) [llm_abstain]
SMILES Cc1cccc(COc2ccccc2C(=O)N(CCCN)Cc2cccs2)c…

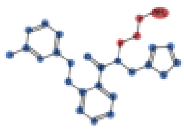
#0 Cc1cccc(CO)c1…



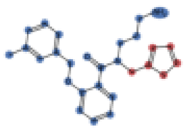
#1 Cc1cccc(CO)c1…



#2 CCCN



#3 Cc1cccs1

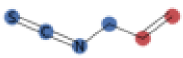


UL9 · Q8R4D5

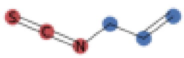
3-isothiocyanatoprop-1-ene

family TRPM8 – cooling/menthol receptor…
organism Mus musculus
PDB 8E4L
fold 4 sites 4 pocket 46.0 res
flavour bitter;spicy;sulfurous [measured]
SMILES C=CCN=C=S

#0 C=C



#1 N=C=S



Catalogue

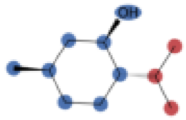
records 205–208 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

XUQ · A0A5S8WF66

(1R,2S,5R)-5-methyl-2-(propan-2-yl)cyclohexan-1-ol

```
family    TRPM8 — cooling/menthol receptor…
organism  Parus major
PDB       9PB6 9ZCQ 9ZEZ
fold 4   sites 12   pocket 68.0 res
flavour   minty;roasted;woody [measured]
SMILES    CC(C)[C@@H]1CC[C@@H](C)C[C@H]1O
```

#0 CCC

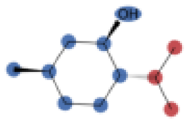


XUQ · Q7Z2W7

(1R,2S,5R)-5-methyl-2-(propan-2-yl)cyclohexan-1-ol

```
family    TRPM8 — cooling/menthol receptor…
organism  Homo sapiens
PDB       9PB5
fold 4   sites 4   pocket 60.0 res
flavour   minty;roasted;woody [measured]
SMILES    CC(C)[C@@H]1CC[C@@H](C)C[C@H]1O
```

#0 CCC

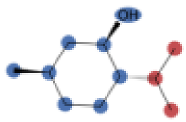


XUQ · Q8R4D5

(1R,2S,5R)-5-methyl-2-(propan-2-yl)cyclohexan-1-ol

```
family    TRPM8 — cooling/menthol receptor…
organism  Mus musculus
PDB       9VJQ
fold 4   sites 4   pocket 53.0 res
flavour   minty;roasted;woody [measured]
SMILES    CC(C)[C@@H]1CC[C@@H](C)C[C@H]1O
```

#0 CCC



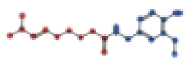
4DY · O35433

(6E)-N-(4-hydroxy-3-methoxybenzyl)-8-methylnon-6-ena…

```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Rattus norvegicus
PDB       7LPA 7LPB 7LPD 7LPE
fold 1   sites 16   pocket 78.5 res
flavour   herbal;odorless [measured]
SMILES    COc1cc(CNC(=O)CCCC/C=C/C(C)C)ccc1O
```

#0 CC(C)C

#1 CC(C)/C=C/CCCCC=…



#2 COc1cc(CN)ccc1O

Catalogue

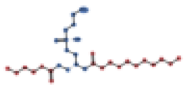
records 209–212 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

65I · O35433;P0CH43

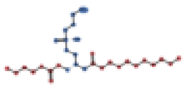
(9R,12R)-15-amino-12-hydroxy-6,12-dioxo-7,11,13-trio…

```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Rattus norvegicus
PDB       7L2R 7L2T
fold 1   sites 8   pocket 54.0 res
flavour   (none) [llm_abstain]
SMILES    CCCCCCCCCCC(=O)O[C@H](COC(=O)CCCCC)CO[P@…
```

#0 CCCCCCCCCCC=O…

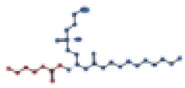


#1 CCCCCCCCCCC=O…



#2 CCCCCC=O

#3 CCCCCC(=O)O

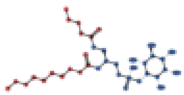


6ES · O35433

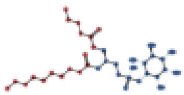
(2S)-1-{[(R)-hydroxy{[(1R,2R,3S,4S,5S,6S)-2,3,4,5,6-…

```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Rattus norvegicus
PDB       5IRZ
fold 1   sites 4   pocket 97.0 res
flavour   (none) [llm_abstain]
SMILES    CCCCCCCCCC(=O)O[C@@H](COC(=O)CCCC)CO[P@@…
```

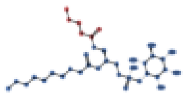
#0 CCCCCCCCCC=O…



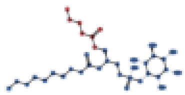
#1 CCCCCCCCCC=O…



#2 CCCCC=O



#3 CCCCC(=O)O



6ET · O35433

capsazepine

```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Rattus norvegicus
PDB       5IS0
fold 1   sites 4   pocket 79.0 res
flavour   (none) [llm_abstain]
SMILES    Oc1cc2c(cc1O)CN(C(=S)NCCc1ccc(Cl)cc1)CCC…
```

#0 S=CNCCc1ccc(Cl)c…

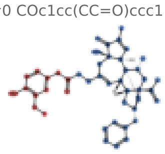
#1 Oc1cc2c(cc1O)CNC…

6EU · O35433

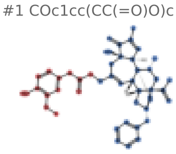
resiniferatoxin

```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Rattus norvegicus
PDB       7L2L 7L2N 7L2O 7L2V 7L2W 7L2X 7MZ5 7MZ7…
fold 1   sites 72   pocket 94.1 res
flavour   spicy [llm]
SMILES    C=C(C)[C@]12C[C@@H](C)[C@@]34O[C@](Cc5cc…
```

#0 COc1cc(CC=O)ccc1…



#1 COc1cc(CC(=O)O)c…



Catalogue

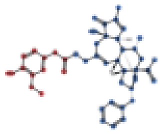
records 213–216 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

6EU · O35433;P0CH43

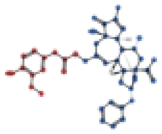
resiniferatoxin

```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Rattus norvegicus
PDB       5IRX 7L2M
fold 1   sites 8   pocket 86.9 res
flavour   spicy [llm]
SMILES    C=C(C)[C@]12C[C@@H](C)[C@@]34O[C@](Cc5cc…
```

#0 COc1cc(CC=O)ccc1…



#1 COc1cc(CC(=O)O)c…



6IY · O35433

(10R,13S)-16-amino-13-hydroxy-7,13-dioxo-8,12,14-tri…

```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Rattus norvegicus
PDB       7L2P 7L2S 7L2W 7L2X 7MZ6
fold 1   sites 20   pocket 59.8 res
flavour   (none) [llm_abstain]
SMILES    CCCCCCCCCCCCCCCC(=O)O[C@H](COC(=O)CCCCCC…
```

#0 CCCCCCCCCCCCCCCC…

#1 CCCCCCC=O

#2 CCCCCCC(=O)O

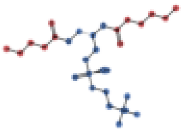
#3 CCCCCCCCCCCCCCCC…

6O8 · O35433;P0CH43

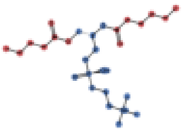
(4R,7S)-4-hydroxy-N,N,N-trimethyl-4,9-dioxo-7-[(pent…

```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Rattus norvegicus
PDB       5IRX
fold 1   sites 4   pocket 72.0 res
flavour   (none) [llm_abstain]
SMILES    CCCCCC(=O)O[C@@H](COC(=O)CCCC)CO[P@](=O)…
```

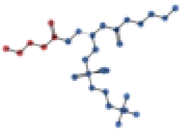
#0 CCCCCC=O CCCCC=O



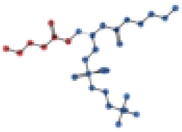
#1 CCCCCC=O…



#2 CCCCC=O



#3 CCCCC(=O)O

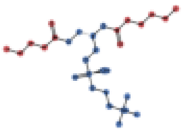


6O8 · O35433

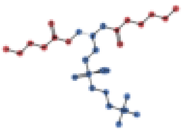
(4R,7S)-4-hydroxy-N,N,N-trimethyl-4,9-dioxo-7-[(pent…

```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Rattus norvegicus
PDB       5IRZ
fold 1   sites 4   pocket 72.0 res
flavour   (none) [llm_abstain]
SMILES    CCCCCC(=O)O[C@@H](COC(=O)CCCC)CO[P@](=O)…
```

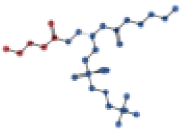
#0 CCCCCC=O CCCCC=O



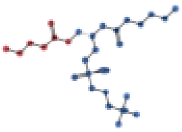
#1 CCCCCC=O…



#2 CCCCC=O



#3 CCCCC(=O)O

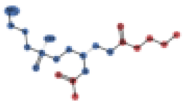


6O9 · O35433;P0CH43

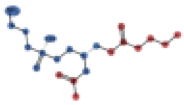
(2S)-2-(acetyloxy)-3-{[(R)-(2-aminoethoxy)(hydroxy)p…

```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Rattus norvegicus
PDB       5IRX
fold 1    sites 4   pocket 36.2 res
flavour   (none) [llm_abstain]
SMILES    CCCCC(=O)OC[C@@H](CO[P@](=O)(O)OCCN)OC(C…
```

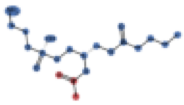
#0 CCCCC=O CC=O



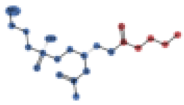
#1 CCCCC(=O)O CC=O



#2 CC=O



#3 CCCCC=O

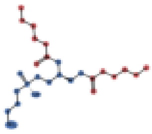


6OE · O35433;P0CH43

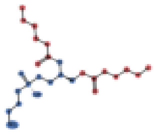
(2S)-3-{[(S)-(2-aminoethoxy)(hydroxy)phosphoryl]oxy}…

```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Rattus norvegicus
PDB       5IRX
fold 1    sites 8   pocket 41.5 res
flavour   (none) [llm_abstain]
SMILES    CCCCCC(=O)OC[C@@H](CO[P@@](=O)(O)OCCN)OC…
```

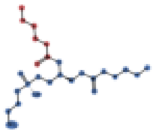
#0 CCCCCC=O…



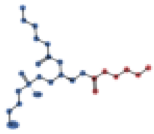
#1 CCCCCC(=O)O…



#2 CCCCCC=O



#3 CCCCCC=O

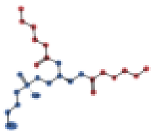


6OE · O35433

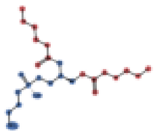
(2S)-3-{[(S)-(2-aminoethoxy)(hydroxy)phosphoryl]oxy}…

```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Rattus norvegicus
PDB       5IRZ
fold 1    sites 16   pocket 38.2 res
flavour   (none) [llm_abstain]
SMILES    CCCCCC(=O)OC[C@@H](CO[P@@](=O)(O)OCCN)OC…
```

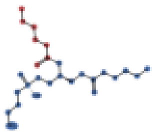
#0 CCCCCC=O…



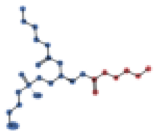
#1 CCCCCC(=O)O…



#2 CCCCCC=O



#3 CCCCCC=O



6OU · Q8NER1

[(2~{R})-1-[2-azanylethoxy(oxidanyl)phosphoryl]oxy-3…

```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Homo sapiens
PDB       8YCP
fold 1    sites 4   pocket 67.5 res
flavour   (none) [llm_abstain]
SMILES    CCCCCCCC/C=C\CCCCCCCC(=O)O[C@H](COC(=O)C…
```

#0 CCCCCCCCC…

#1 CCCCCCCCC…

#2 CCCCCCCCC…

#3 CCCCCCCCC…

Catalogue

records 221–224 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

6OU · O35433

[(2~{R})-1-[2-azanylethoxy(oxidanyl)phosphoryl]oxy-3...

family TRPV1 — capsaicin/vanilloid receptor…
organism Rattus norvegicus
PDB 7LP9 7LPA 7LPB 7LPC 7LPD 7LPE 7RQU 7RQV…
fold 1 sites 160 pocket 46.0 res
flavour (none) [llm_abstain]
SMILES CCCCCCCC/C=C\CCCCCCCC(=O)O[C@H](COC(=O)C…

#0 CCCCCCCCC...

#1 CCCCCCCCC...

#2 CCCCCCCCC...

#3 CCCCCCCCC...

A1C8H · Q8NER1

4-[3-(trifluoromethyl)pyridin-2-yl]-N-[5-(trifluorom...

family TRPV1 — capsaicin/vanilloid receptor…
organism Homo sapiens
PDB 1ICO
fold 1 sites 4 pocket 76.0 res
flavour (none) [llm_abstain]
SMILES O=C(Nc1ccc(C(F)(F)F)cn1)N1CCN(c2ncccc2C(…

#0 FC(F)(F)c1cccnc1 ●

#1 Nc1ccc(C(F)(F)F)... ●

#2 O=CNc1ccc(C(F)(F... ●

#3 FC(F)(F)c1cccnc1... ●

A1C8I · Q8NER1

2-[(2P)-2-(2-{(E)-2-[4-(trifluoromethyl)phenyl]ethen...

family TRPV1 — capsaicin/vanilloid receptor…
organism Homo sapiens
PDB 1ICN
fold 1 sites 4 pocket 84.0 res
flavour (none) [llm_abstain]
SMILES CC(C)(O)c1ccccc1-c1ccc2[nH]c(/C=C/c3ccc(…

#0 CC(C)(O)c1ccccc1

#1 Cc1ccc(C(F)(F)F)...

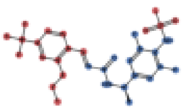
#2 Cc1nc2cc(-c3cccc... ●

A1C8J · Q8NER1

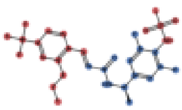
(2E)-N-{(1R)-1-[3,5-difluoro-4-(methanesulfonamido)p...

family TRPV1 — capsaicin/vanilloid receptor…
organism Homo sapiens
PDB 1ICL
fold 1 sites 4 pocket 75.0 res
flavour (none) [llm_abstain]
SMILES CCCc1nc(C(F)(F)F)ccc1/C=C/C(=O)N[C@H](C)…

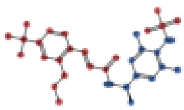
#0 Cc1ccc(C(F)(F)F)... ●



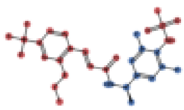
#1 Cc1ccc(C(F)(F)F)... ●



#2 CCCc1nc(C(F)(F)F... ●



#3 CCCc1nc(C(F)(F)F... ●



Catalogue

records 225–228 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

A1C8K · Q8NER1

N-[(4-hydroxy-2-iodo-5-methoxyphenyl)methyl]-8-methy…

```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Homo sapiens
PDB       11CK
fold 1   sites 4   pocket 80.0 res
flavour   (none) [llm_abstain]
SMILES    COc1cc(CNC(=O)CCCCCCC(C)C)c(I)cc1O
```

#0 CC(C)CCCCCCC=O

#1 COc1cc(CN)c(I)cc… ●

A1CHV · Q8NER1

3-(1-benzothiophen-2-yl)-N-[(4-hydroxy-3-methoxyphen…

```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Homo sapiens
PDB       9PEA
fold 1   sites 4   pocket 78.0 res
flavour   (none) [llm_abstain]
SMILES    COc1cc(CNC(=O)CCc2cc3ccccc3s2)ccc1O
```

#0 COc1cc(CN)ccc1O

#1 O=CCCc1cc2ccccc2… ●

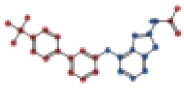
A1D6R · O35433

~{N}-[4-[6-[4-(trifluoromethyl)phenyl]pyrimidin-4-yl…

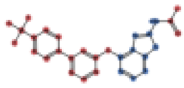
```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Escherichia coli K-12
PDB       9W4M
fold 1   sites 4   pocket 80.2 res
flavour   (none) [llm_abstain]
SMILES    CC(=O)Nc1nc2c(Oc3cc(-c4ccc(C(F)(F)F)cc4)…
```

#0 CC=O…

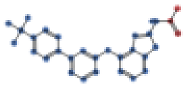
#1 CC=O… ●



#2 CC=O… ●



#3 CC=O

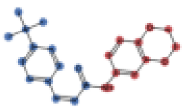


A1D6W · O35433

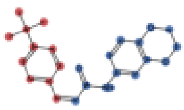
(~{Z})-3-(4-~{tert}-butylphenyl)-~{N}-(2,3-dihydro-1…

```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Escherichia coli K-12
PDB       9W4T
fold 1   sites 4   pocket 84.0 res
flavour   (none) [llm_abstain]
SMILES    CC(C)(C)c1ccc(/C=C\C(=O)Nc2ccc3c(c2)OCCO…
```

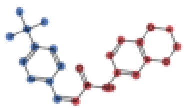
#0 Nc1ccc2c(c1)OCCO…



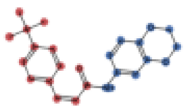
#1 Cc1ccc(C(C)(C)C)…



#2 CC(=O)Nc1ccc2c(c…



#3 CC(C)(C)c1ccc(/C… ●



Catalogue

records 229–232 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

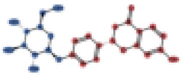
A1ETR · Q8NER1

Liquiritin

```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Homo sapiens
PDB       9VTZ
fold 1   sites 1   pocket 75.0 res
flavour   (none) [llm_abstain]
SMILES    O=C1C[C@@H](c2ccc(O[C@@H]3O[C@H](CO)[C@@…
```

#0 OC[C@H]1O[C@@H](…

#1 O=C1C[C@@H](c2cc…

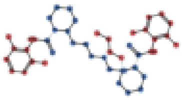


A1LZ2 · Q8NER1

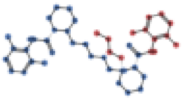
(1~{R},2~{S})-1-butyl-~{N}-(2,6-dimethylphenyl)-1-[5…

```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Homo sapiens
PDB       8YCP
fold 1   sites 4   pocket 95.8 res
flavour   (none) [llm_abstain]
SMILES    CCCC[N@@+]1(CCCCCN2CCCC[C@H]2C(=O)Nc2c(C…
```

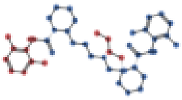
#0 CCCC…



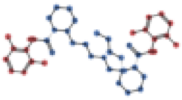
#1 CCCC…



#2 CCCC…



#3 Cc1cccc(C)c1N…



DU0 · Q8NER1

2-[2-[(1~{S},2~{S},4~{S},5'~{R},6~{R},7~{S},8~{R},9~…

```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Homo sapiens
PDB       8GFA
fold 1   sites 4   pocket 45.0 res
flavour   (none) [llm_abstain]
SMILES    C[C@@H]1CC[C@@]2(OC1)O[C@H]1C[C@H]3[C@@H…
```

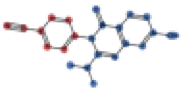
#0 CCC(CO)CO

EZI · Q8NER1

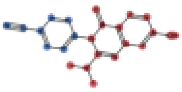
4-(7-Hydroxy-2-isopropyl-4-oxoquinazolin-3(4H)-yl)be…

```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Homo sapiens
PDB       8JQR 8X94
fold 1   sites 8   pocket 68.8 res
flavour   (none) [llm_abstain]
SMILES    CC(C)c1nc2cc(O)ccc2c(=O)n1-c1ccc(C#N)cc1
```

#0 N#Cc1ccccc1



#1 CC(C)c1nc2cc(O)c… ●



Catalogue

records 233–236 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

LFA · Q8NER1

EICOSANE

```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Homo sapiens
PDB       11CL
fold 1   sites 4   pocket 54.0 res
flavour   fatty [measured]
SMILES    CCCCCCCCCCCCCCCCCCCC
```

#0 CC

#1 CC

#2 CCC

#3 CCC

NKN · O35433

(2R)-2-hydroxy-3-(phosphonooxy)propyl tetradecanoate

```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Rattus norvegicus
PDB       8T0C 8T0Y 8T10 8T3L 8T3M
fold 1   sites 12   pocket 89.2 res
flavour   (none) [llm_abstain]
SMILES    CCCCCCCCCCCCCC(=O)OC[C@@H](O)COP(=O)(O)O
```

#0 C[C@@H](O)COP(=O…

#1 O=P(O)(O)OC[C@H]…

#2 CCCCCCCCCCCCCC=O

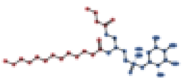
#3 CCCCCCCCCCCCCC(=…

XJ7 · O35433

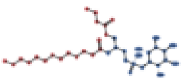
(2S)-1-(butanoyloxy)-3-{[(R)-hydroxy{[(1r,2R,3S,4S,5…

```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Rattus norvegicus
PDB       7L2H 7L2J 7L2P 7L2S 7MZ6 7MZ9 7MZA 7MZB…
fold 1   sites 28   pocket 98.7 res
flavour   (none) [llm_abstain]
SMILES    CCCCCCCCCCCCC(=O)O[C@@H](COC(=O)CCC)CO[P…
```

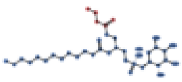
#0 CCCCCCCCCCCCC=O…



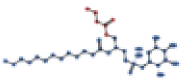
#1 CCCCCCCCCCCCC=O…



#2 CCCC=O



#3 CCCC(=O)O

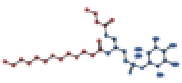


XJ7 · O35433;P0CH43

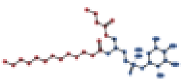
(2S)-1-(butanoyloxy)-3-{[(R)-hydroxy{[(1r,2R,3S,4S,5…

```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Rattus norvegicus
PDB       7L2R 7L2T 7L2U
fold 1   sites 12   pocket 95.9 res
flavour   (none) [llm_abstain]
SMILES    CCCCCCCCCCCCC(=O)O[C@@H](COC(=O)CCC)CO[P…
```

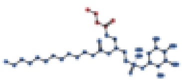
#0 CCCCCCCCCCCCC=O…



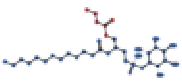
#1 CCCCCCCCCCCCC=O…



#2 CCCC=O



#3 CCCC(=O)O



Catalogue

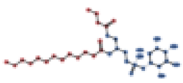
records 237–240 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

XJA · O35433

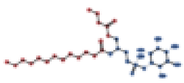
(2S)-1-(butanoyloxy)-3-{[(R)-hydroxy{[(1S,2R,3R,4R,5…

```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Rattus norvegicus
PDB       7L2I
fold 1   sites 4   pocket 98.8 res
flavour   (none) [llm_abstain]
SMILES    CCCCCCCCCCCCC(=O)O[C@@H](COC(=O)CCC)CO[P…
```

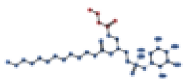
#0 CCCCCCCCCCCCC=O…



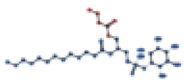
#1 CCCCCCCCCCCCC=O…



#2 CCCC=O



#3 CCCC(=O)O



XJD · O35433

(10R,13S)-16-amino-13-hydroxy-7,13-dioxo-8,12,14-tri…

```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Rattus norvegicus
PDB       7L2H
fold 1   sites 4   pocket 52.0 res
flavour   (none) [llm_abstain]
SMILES    CCCCCCCCCCCCC(=O)O[C@H](COC(=O)CCCCCC)CO…
```

#0 CCCCCCCCCCCCC=O…

#1 CCCCCCCCCCCCC=O…

#2 CCCCCCC=O

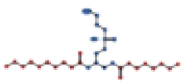
#3 CCCCCCC(=O)O

XKP · O35433;P0CH43

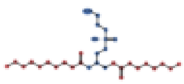
(11R,14S)-17-amino-14-hydroxy-8,14-dioxo-9,13,15-tri…

```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Rattus norvegicus
PDB       7L2U
fold 1   sites 4   pocket 48.2 res
flavour   (none) [llm_abstain]
SMILES    CCCCCCCCCC(=O)O[C@H](COC(=O)CCCCCCC)CO[P…
```

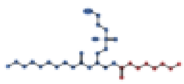
#0 CCCCCCCCCC=O…



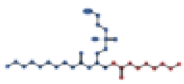
#1 CCCCCCCCCC=O…



#2 CCCCCCCC=O



#3 CCCCCCCC(=O)O

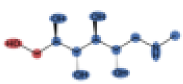


XPJ · O35433

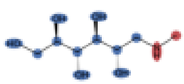
1-deoxy-1-(methylamino)-D-allitol

```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Rattus norvegicus
PDB       7L2X
fold 1   sites 1   pocket 53.0 res
flavour   (none) [llm_abstain]
SMILES    CNC[C@H](O)[C@H](O)[C@H](O)[C@H](O)CO
```

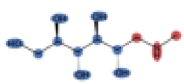
#0 CO



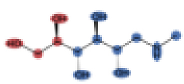
#1 CN



#2 CNC



#3 OCCO



Catalogue

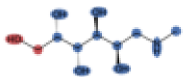
records 241–243 of 243 · core in blue, R-groups in red, ● = R-group absent from the pretraining vocabulary

XPS · O35433

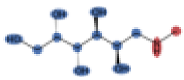
6-deoxy-6-(methylamino)-D-galactitol

```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Rattus norvegicus
PDB       7L2V
fold 1   sites 1   pocket 43.0 res
flavour   (none) [llm_abstain]
SMILES    CNC[C@@H](O)[C@H](O)[C@H](O)[C@@H](O)CO
```

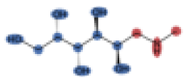
#0 CO



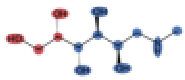
#1 CN



#2 CNC



#3 OCCO

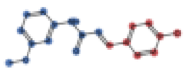


ZEI · Q8NER1

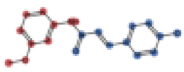
(2E)-3-(4-chlorophenyl)-N-(3-methoxyphenyl)prop-2-en…

```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Homo sapiens
PDB       8GFA
fold 1   sites 4   pocket 93.0 res
flavour   (none) [llm_abstain]
SMILES    COc1cccc(NC(=O)/C=C/c2ccc(Cl)cc2)c1
```

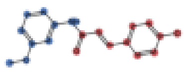
#0 Cc1ccc(Cl)cc1



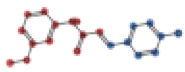
#1 COc1cccc(N)c1



#2 O=C/C=C/c1ccc(Cl…



#3 CC(=O)Nc1cccc(OC…

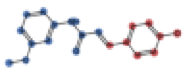


ZEI · O35433

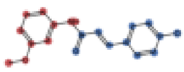
(2E)-3-(4-chlorophenyl)-N-(3-methoxyphenyl)prop-2-en…

```
family    TRPV1 — capsaicin/vanilloid receptor…
organism  Rattus norvegicus
PDB       9W4N
fold 1   sites 4   pocket 80.0 res
flavour   (none) [llm_abstain]
SMILES    COc1cccc(NC(=O)/C=C/c2ccc(Cl)cc2)c1
```

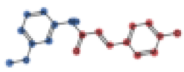
#0 Cc1ccc(Cl)cc1



#1 COc1cccc(N)c1



#2 O=C/C=C/c1ccc(Cl…



#3 CC(=O)Nc1cccc(OC…